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Isolated gate driver, traction inverter, and electric vehicle

Abstract

An isolated gate driver has: a switch connection terminal configured to have a switching element externally connected to it; a non-volatile memory having adjustment data written to it; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive the gate of the switching element with a characteristic set based on a value stored in the register; and a control logic circuit configured to keep the gate driving circuit in a disabled state until completion of reading the adjustment data from the non-volatile memory and storing it in the register.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This nonprovisional application is a continuation application of International Patent Application No. PCT/JP2022/002153 filed on Jan. 21, 2022, which claims priority Japanese Patent Application No. 2021-042136 filed on Mar. 16, 2021, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The invention disclosed herein relates to isolated gate drivers, and also relates to traction inverters and electric vehicles that employ isolated gate drivers.

BACKGROUND ART

(2) Isolated gate drivers are used in a variety of sets (e.g., traction inverters for electric vehicles).

(3) One example of known technology related to what is mentioned above is found in Patent Document 1 identified below.

CITATION LIST

Patent Literature

(4) Patent Document 1: WO 2011/055611

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating the basic configuration of a signal transmission device.

(2) FIG. 2 is a diagram illustrating the basic structure of a transformer chip.

(3) FIG. 3 is a perspective view of a semiconductor device used as a two-channel transformer chip.

(4) FIG. 4 is a plan view of the semiconductor device shown in FIG. 3.

(5) FIG. 5 is a plan view of a layer in the semiconductor device shown in FIG. 3 where low-potential coils are formed.

(6) FIG. 6 is a plan view of a layer in the semiconductor device shown in FIG. 3 where high-potential coils are formed.

(7) FIG. 7 is a cross-sectional view taken along line VIII-VIII shown in FIG. 6.

(8) FIG. 8 is an enlarged view (showing a separation structure) of region XIII shown in FIG. 7.

(9) FIG. 9 is a diagram schematically showing an example of the layout of a transformer chip.

(10) FIG. 10 is a diagram showing the basic configuration of a traction inverter.

(11) FIG. 11 is a diagram showing a novel embodiment of an isolated gate driver.

(12) FIG. 12 is a diagram showing a start-up sequence of an isolated gate driver.

(13) FIG. 13 is a diagram showing a first implementation example of a non-volatile memory.

(14) FIG. 14 is a diagram showing a second implementation example of a non-volatile memory.

(15) FIG. 15 is a diagram showing a third implementation example of a non-volatile memory.

(16) FIG. 16 is a diagram showing a first operation example of fault output.

(17) FIG. 17 is a diagram showing a second operation example of fault output.

(18) FIG. 18 is a diagram showing a third operation example of fault output.

(19) FIG. 19 is a diagram showing a fourth operation example of fault output.

(20) FIG. 20 is a diagram showing a first example of sharing a fault signal output terminal.

(21) FIG. 21 is a diagram showing a second example of sharing a fault signal output terminal.

(22) FIG. 22 is a diagram showing a third example of sharing a fault signal output terminal.

(23) FIG. 23 is a diagram showing the exterior appearance of an electric vehicle.

DESCRIPTION OF EMBODIMENTS

Signal Transmission Device (Basic Configuration)

(24) FIG. 1 is a diagram illustrating the basic configuration of a signal transmission device. The signal transmission device **200** of this configuration example is a semiconductor integrated circuit device (what is generally called an isolated gate driver IC) that, while isolating between a primary circuit system **200p** (VCC1-GND1 system) and a secondary circuit system **200s** (VCC2-GND2 system), transmits a pulse signal from the primary circuit system **200p** to the secondary circuit system **200s** to drive the gate of a switching device (unillustrated) provided in the secondary circuit system **200s**. The signal transmission device **200** has, for example, a controller chip **210**, a driver chip **220**, and a transformer chip **230** sealed in a single package.

(25) The controller chip **210** is a semiconductor chip that operates by being supplied with a supply voltage VCC1 (e.g., seven volts at the maximum with respect to GND1). The controller chip **210** has, for example, a pulse transmission circuit **211** and buffers **212** and **213** integrated in it.

(26) The pulse transmission circuit **211** is a pulse generator that generates transmission pulse signals **S11** and **S21** according to an input pulse signal IN. More specifically, when indicating that the input pulse signal IN is at high level, the pulse transmission circuit **211** pulse-drives (outputs a single or a plurality of pulses in) the transmission pulse signal **S11**; when indicating that the input pulse signal IN is at low level, the pulse transmission circuit **211** pulse-drives the transmission pulse signal **S21**. That is, the pulse transmission circuit **211** pulse-drives either the transmission pulse signal **S11** or **S21** according to the logic level of the input pulse signal IN.

(27) The buffer **212** receives the transmission pulse signal **S11** from the pulse transmission circuit **211**, and pulse-drives the transformer chip **230** (more specifically, a transformer **231**).

(28) The buffer **213** receives the transmission pulse signal **S21** from the pulse transmission circuit **211**, and pulse-drives the transformer chip **230** (more specifically, a transformer **232**).

(29) The driver chip **220** is a semiconductor chip that operates by being supplied with a supply voltage VCC2 (e.g., 30 volts at the maximum with respect to GND2). The driver chip **220** has, for example, buffers **221** and **222**, a pulse reception circuit **223**, and a driver **224** integrated in it.

(30) The buffer **221** performs waveform shaping on a reception pulse signal **S12** induced in the transformer chip **230** (specifically, the transformer **231**), and outputs the result to the pulse reception circuit **223**.

(31) The buffer **222** performs waveform shaping on a reception pulse signal **S22** induced in the transformer chip **230** (specifically, the transformer **232**), and outputs the result to the pulse reception circuit **223**.

(32) According to the reception pulse signals **S12** and **S22** fed to it via the buffers **221** and **222**, the pulse reception circuit **223** drives the driver **224** to generate an output pulse signal OUT. More specifically, the pulse reception circuit **223** drives the driver **224** to raise the output pulse signal OUT to high level in response to the reception pulse signal **S12** being pulse-driven and to drop the output pulse signal OUT to low level in response to the reception pulse signal **S22** being pulse-driven. That is, the pulse reception circuit **223** switches the logic level of the output pulse signal OUT according to the logic level of the input pulse signal IN. As the pulse reception circuit **223**, for example, an RS flip-flop can be suitably used.

(33) The driver **224** generates the output pulse signal OUT under the driving and control of the pulse reception circuit **223**.

(34) The transformer chip **230**, while isolating between the controller chip **210** and the driver chip **220** on a direct-current basis using the transformers **231** and **232**, outputs the transmission pulse signals **S11** and **S21** fed to the transformer chip **230** from the pulse transmission circuit **211** to, as the reception pulse signals **S12** and **S22**, the pulse reception circuit **223**. In the present description, “isolating on a direct-current basis” means leaving two elements to be isolated from each other unconnected by a conductor.

(35) More specifically, the transformer **231** outputs, according to the transmission pulse signal **S11** fed to the primary coil **231p**, the reception pulse signal **S12** from the secondary coil **231s**.

Likewise, the transformer **232** outputs, according to the transmission pulse signal **S21** fed to the primary coil **232p**, the reception pulse signal **S22** from the secondary coil **232s**.

(36) In this way, owing to the characteristics of spiral coils used in isolated communication, the input pulse signal IN is split into two transmission pulse signals **S11** and **S21** (corresponding to a rise signal and a fall signal) to be transmitted via the two transformers **231** and **232** from the primary circuit system **200p** to the secondary circuit system **200s**.

(37) Note that the signal transmission device **200** of this configuration example has, separately from the controller chip **210** and the driver chip **220**, the transformer chip **230** that incorporates the transformers **231** and **232** alone, and those three chips are sealed in a single package.

(38) With this configuration, the controller chip **210** and the driver chip **220** can each be formed by a common low- to middle-withstand-voltage process (with a withstand voltage of several volts to several tens of volts). This eliminates the need for a dedicated high-withstand-voltage process (with a withstand voltage of several kilovolts), and helps reduce manufacturing costs.

(39) The signal transmission device **200** can be employed suitably, for example, in a power supply device or motor driving device in a vehicle-mounted device incorporated in a vehicle. Such a vehicle can be an engine vehicle or an electric vehicle (an xEV such as a BEV [battery electric vehicle], HEV [hybrid electric vehicle], PHEV/PHV [plug-in hybrid electric vehicle/plug-in hybrid vehicle], or FCEV/FCV [fuel cell electric vehicle/fuel cell vehicle]).

Transformer Chip (Basic Structure)

(40) Next, the basic structure of the transformer chip **230** will be described. FIG. **2** is a diagram showing the basic structure of the transformer chip **230**. In the transformer chip **230** shown there, the transformer **231** includes a primary coil **231p** and a secondary coil **231s** that face each other in the up-down direction; the transformer **232** includes a primary coil **232p** and a secondary coil **232s** that face each other in the up-down direction.

(41) The primary coils **231p** and **232p** are both formed in a first wiring layer (lower layer) **230a** in the transformer chip **230**. The secondary coils **231s** and **231s** are both formed in a second wiring layer (the upper layer in the diagram) **230b** in the transformer chip **230**. The secondary coil **231s** is disposed right above the primary coil **231p** and faces the primary coil **231p**; the secondary coil **232s** is disposed right above the primary coil **232p** and faces the primary coil **232p**.

(42) The primary coil **231p** is laid in a spiral shape so as to encircle an internal terminal **X21** clockwise, starting at the first terminal of the primary coil **231p**, which is connected to the internal terminal **X21**. The second terminal of the primary coil **231p**, which corresponds to its end point, is connected to an internal terminal **X22**. Likewise, the primary coil **232p** is laid in a spiral shape so as to encircle an internal terminal **X23** anticlockwise, starting at the first terminal of the primary coil **232p**, which is connected to the internal terminal **X23**. The second terminal of the primary coil **232p**, which corresponds to its end point, is connected to the internal terminal **X22**. The internal terminals **X21**, **X22**, and **X23** are arrayed on a straight line in the illustrated order.

(43) The internal terminal **X21** is connected, via a wiring **Y21** and a via **Z21** both conductive, to an external terminal **T21** in the second layer **230b**. The internal terminal **X22** is connected, via a wiring **Y22** and a via **Z22** both conductive, to an external terminal **T22** in the second layer **230b**. The internal terminal **X23** is connected, via a wiring **Y23** and a via **Z23** both conductive, to an external terminal **T23** in the second layer **230b**. The external terminals **T21** to **T23** are disposed in a straight row and are used for wire-bonding with the controller chip **210**.

(44) The secondary coil **231s** is laid in a spiral shape so as to encircle an external terminal **T24** anticlockwise, starting at the first terminal of the secondary coil **231s**, which is connected to the external terminal **T24**. The second terminal of the secondary coil **231s**, which corresponds to its end point, is connected to an external terminal **T25**. Likewise, the secondary coil **232s** is laid in a spiral shape so as to encircle an external terminal **T26** clockwise, starting at the first terminal of the secondary coil **232s**, which is connected to the external terminal **T26**. The second terminal of the secondary coil **232s**, which corresponds to its end point, is connected to the external terminal **T25**. The external terminals **T24**, **T25**, and **T26** are disposed in a straight row in the illustrated order and are used for wire-bonding with the driver chip **220**.

(45) The secondary coils **231s** and **232s** are AC-connected to the primary coils **231p** and **232p**, respectively, by magnetic coupling, and are DC-isolated from the primary coils **231p** and **232p**. That is, the driver chip **220** is AC-connected to the controller chip **210** via the transformer chip **230**, and is DC-isolated from the controller chip **210** by the transformer chip **230**.

Transformer Chip (Two-Channel Type)

(46) FIG. **3** is a perspective view of a semiconductor device **5** used as a two-channel transformer chip. FIG. **4** is a plan view of the semiconductor device **5** shown in FIG. **3**. FIG. **5** is a plan view

showing a layer in the semiconductor device **5** shown in FIG. **3** where low-potential coils **22** (corresponding to the primary coils of transformers) are formed. FIG. **6** is a plan view showing a layer in the semiconductor device **5** shown in FIG. **3** where high-potential coils **23** (corresponding to the secondary coils of transformers) are formed. FIG. **7** is a sectional view along line VIII-VIII shown in FIG. **6**. FIG. **8** is an enlarged view of region XIII shown in FIG. **7**, showing a separation structure **130**.

(47) Referring to FIGS. **3** to **7**, the semiconductor device **5** includes a semiconductor chip **41** in the shape of a rectangular parallelepiped. The semiconductor chip **41** contains at least one of silicon, a wide band gap semiconductor, and a compound semiconductor.

(48) The wide band gap semiconductor is a semiconductor with a band gap larger than that of silicon (about 1.12 eV). Preferably, the wide band gap semiconductor has a band gap of 2.0 eV or more. The wide band gap semiconductor can be SiC (silicon carbide). The compound semiconductor can be a III-V group compound semiconductor. The compound semiconductor can contain at least one of aluminum nitride (AlN), indium nitride (InN), gallium nitride (GaN), and gallium arsenide (GaAs).

(49) In the embodiment, the semiconductor chip **41** includes a semiconductor substrate made of silicon. The semiconductor chip **41** can be an epitaxial substrate that has a stacked structure composed of a semiconductor substrate made of silicon and an epitaxial layer made of silicon. The semiconductor substrate can be of an n-type or p-type conductivity. The epitaxial layer can be of an n-type or p-type.

(50) The semiconductor chip **41** has a first principal surface **42** at one side, a second principal surface **43** at the other side, and chip side walls **44A** to **44D** that connect the first and second principal surfaces **42** and **43** together. As seen in a plan view from the normal direction Z to them (hereinafter simply expressed as “as seen in a plan view”), the first and second principal surfaces **42** and **43** are each formed in a quadrangular shape (in the embodiment, in a rectangular shape).

(51) The chip side walls **44A** to **44D** includes a first chip side wall **44A**, a second chip side wall **44B**, a third chip side wall **44C**, and a fourth chip side wall **44D**. The first and second chip side walls **44A** and **44B** constitute the longer sides of the semiconductor chip **41**. The first and second chip side walls **44A** and **44B** extend along a first direction X and face away from each other in a second direction Y. The third and fourth chip side walls **44C** and **44D** constitute the shorter sides of the semiconductor chip **41**. The third and fourth chip side walls **44C** and **44D** extend in the second direction Y and face away from each other in the first direction X. The chip side walls **44A** to **44D** have polished surfaces.

(52) The semiconductor device **5** further includes an insulation layer **51** formed on the first principal surface **42** of the semiconductor chip **41**. The insulation layer **51** has an insulation principal surface **52** and insulation side walls **53A** to **53D**. The insulation principal surface **52** is formed in a quadrangular shape (in the embodiment, a rectangular shape) that fits the first principal surface **42** as seen in a plan view. The insulation principal surface **52** extends parallel to the first principal surface **42**.

(53) The insulation side walls **53A** to **53D** include a first insulation side wall **53A**, a second insulation side wall **53B**, a third insulation side wall **53C**, and a fourth insulation side wall **53D**. The insulation side walls **53A** to **53D** extend from the circumferential edge of the insulation principal surface **52** toward the semiconductor chip **41**, and are continuous with the chip side walls **44A** to **44D**. Specifically, the insulation side walls **53A** to **53D** are formed to be flush with the chip side walls **44A** to **44D**. The insulation side walls **53A** to **53D** constitute polished surfaces that are flush with the chip side walls **44A** to **44D**.

(54) The insulation layer **51** has a stacked structure of multilayer insulation layers that include a bottom insulation layer **55**, a top insulation layer **56**, and a plurality of (in the embodiment, eleven) interlayer insulation layers **57**. The bottom insulation layer **55** is an insulation layer that directly covers the first principal surface **42**. The top insulation layer **56** is an insulation layer that

constitutes the insulation principal surface **52**. The plurality of interlayer insulation layers **57** are insulation layers that are interposed between the bottom and top insulation layers **55** and **56**. In the embodiment, the bottom insulation layer **55** has a single-layer structure that contains silicon oxide. In the embodiment, the top insulation layer **56** has a single-layer structure that contains silicon oxide. The bottom and top insulation layers **55** and **56** can each have a thickness of 1 μm or more but 3 μm or less (e.g., about 2 μm).

(55) The plurality of interlayer insulation layers **57** each have a stacked structure that includes a first insulation layer **58** at the bottom insulation layer **55** side and a second insulation layer **59** at the top insulation layer **56** side. The first insulation layer **58** can contain silicon nitride. The first insulation layer **58** is formed as an etching stopper layer for the second insulation layer **59**. The first insulation layer **58** can have a thickness of μm or more but 1 μm or less (e.g., about 0.3 μm).

(56) The second insulation layer **59** is formed on top of the first insulation layer **58**, and contains an insulating material different from that of the first insulation layer **58**. The second insulation layer **59** can contain silicon oxide. The second insulation layer **59** can have a thickness of 1 μm or more but 3 μm or less (e.g., about 2 μm). Preferably, the second insulation layer **59** is given a thickness larger than that of the first insulation layer **58**.

(57) The insulation layer **51** can have a total thickness DT of 5 μm or more but 50 μm or less. The insulation layer **51** can have any total thickness DT and any number of interlayer insulation layers **57** stacked together, which are adjusted according to the desired dielectric strength voltage (dielectric breakdown withstand voltage). The bottom insulation layer **55**, the top insulation layer **56**, and the interlayer insulation layers **57** can employ any insulating material, which is thus not limited to any particular insulating material.

(58) The semiconductor device **5** includes a first functional device **45** formed in the insulation layer **51**. The first functional device **45** includes one or a plurality of (in the embodiment, a plurality of) transformers **21** (corresponding the transformers mentioned previously). That is, the semiconductor device **5** is a multichannel device that includes a plurality of transformers **21**. The plurality of transformers **21** are formed in an inner part of the insulation layer **51**, at intervals from the insulation side walls **53A** to **53D**. The plurality of transformers **21** are formed at intervals from each other in the first direction X.

(59) Specifically, the plurality of transformers **21** include a first transformer **21A**, a second transformer **21B**, a third transformer **21C**, and a fourth transformer **21D** that are formed in this order from the insulation side wall **53C** side to the insulation side wall **53D** side as seen in a plan view. The plurality of transformers **21A** to **21D** have similar structures. In the following description, the structure of the first transformer **21A** will be described as an example. No separate description will be given of the structures of the second, third, and fourth transformers **21B**, **21C**, and **21D**, to which the description of the structure of the first transformer **21A** is to be taken to apply.

(60) Referring to FIGS. 5 to 7, the first transformer **21A** includes a low-potential coil **22** and a high-potential coil **23**. The low-potential coil **22** is formed in the insulation layer **51**. The high-potential coil **23** is formed in the insulation layer **51** so as to face the low-potential coil **22** in the normal direction Z. In the embodiment, the low- and high-potential coils **22** and **23** are formed in a region between the bottom and top insulation layers **55** and **56** (i.e., in the plurality of interlayer insulation layer **57**).

(61) The low-potential coil **22** is formed in the insulation layer **51**, at the bottom insulation layer **55** (semiconductor chip **41**) side, and the high-potential coil **23** is formed in the insulation layer **51**, at the top insulation layer **56** (insulation principal surface **52**) side with respect to the low-potential coil **22**. That is, the high-potential coil **23** faces the semiconductor chip **41** across the low-potential coil **22**. The low- and high-potential coils **22** and **23** can be disposed at any places. The high-potential coil **23** can face the low-potential coil **22** across one or more interlayer insulation layers **57**.

(62) The distance between the low- and high-potential coils **22** and **23** (i.e., the number of interlayer insulation layers **57** stacked together) is adjusted appropriately according to the dielectric strength voltage and electric field strength between the low- and high-potential coils **22** and **23**. In the embodiment, the low-potential coil **22** is formed in the third interlayer insulation layer **57** as counted from the bottom insulation layer **55** side. In the embodiment, the high-potential coil **23** is formed in the first interlayer insulation layer **57** as counted from the top insulation layer **56** side.

(63) The low-potential coil **22** is embedded in the interlayer insulation layer **57** so as to penetrate the first and second insulation layers **58** and **59**. The low-potential coil **22** includes a first inner end **24**, a first outer end **25**, and a first spiral portion **26** that is patterned in a spiral shape between the first inner and outer ends **24** and **25**. The first spiral portion **26** is patterned in a spiral shape that extends in an elliptical (oval) shape as seen in a plan view. The part of the first spiral portion **26** that forms its inner circumferential edge defines a first inner region **66** that is in an elliptical shape as seen in a plan view.

(64) The first spiral portion **26** can have a number of turns of 5 or more but 30 or less. The first spiral portion **26** can have a width of 0.1 μm or more but 5 μm or less. Preferably, the first spiral portion **26** has a width of 1 μm or more but 3 μm or less. The width of the first spiral portion **26** is defined by its width in the direction orthogonal to the spiraling direction. The first spiral portion **26** has a first winding pitch of 0.1 μm or more but 5 μm or less. Preferably, the first winding pitch is 1 μm or more but 3 μm or less. The first winding pitch is defined by the distance between two parts of the first spiral portion **26** that are adjacent to each other in the direction orthogonal to the spiraling direction.

(65) The first spiral portion **26** can have any winding shape and the first inner region **66** can have any planar shape, which are thus not limited to those shown in FIG. 5 etc. The first spiral portion **26** can be wound in a polygonal shape, such as a triangular or quadrangular shape, or in a circular shape as seen in a plan view. The first inner region **66** can be defined, so as to fit the winding shape of the first spiral portion **26**, in a polygonal shape, such as a triangular or quadrangular shape, or in a circular shape as seen in a plan view.

(66) The low-potential coil **22** can contain at least one of titanium, titanium nitride, copper, aluminum, and tungsten. The low-potential coil **22** can have a stacked structure composed of a barrier layer and a body layer. The barrier layer defines a recessed space in the interlayer insulation layer **57**. The barrier layer can contain at least one of titanium and titanium nitride. The body layer can contain at least one of copper, aluminum, and tungsten.

(67) The high-potential coil **23** is embedded in the interlayer insulation layer **57** so as to penetrate the first and second insulation layers **58** and **59**. The high-potential coil **23** includes a second inner end **27**, a second outer end **28**, and a second spiral portion **29** that is patterned in a spiral shape between the second inner and outer ends **27** and **28**. The second spiral portion **29** is patterned in a spiral shape that extends in an elliptical (oval) shape as seen in a plan view. The part of the second spiral portion **29** that forms its inner circumferential edge defines a second inner region **67** that is in an elliptical shape as seen in a plan view in the embodiment. The second inner region **67** in the second spiral portion **29** faces the first inner region **66** in the first spiral portion **26** in the normal direction Z.

(68) The second spiral portion **29** can have a number of turns of 5 or more but 30 or less. The number of turns of the second spiral portion **29** relative to that of the first spiral portion **26** is adjusted according to the target value of voltage boosting. Preferably, the number of turns of the second spiral portion **29** is larger than that of the first spiral portion **26**. Needless to say, the number of turns of the second spiral portion **29** can be smaller than or equal to that of the first spiral portion **26**.

(69) The second spiral portion **29** can have a width of 0.1 μm or more but 5 μm or less. Preferably, the second spiral portion **29** has a width of 1 μm or more but 3 μm or less. The width of the second spiral portion **29** is defined by its width in the direction orthogonal to the spiraling direction.

Preferably, the width of the second spiral portion **29** is equal to the width of the first spiral portion **26**.

(70) The second spiral portion **29** can have a second winding pitch of 0.1 μm or more but 5 μm or less. Preferably, the second winding pitch is 1 μm or more but 3 μm or less. The second winding pitch is defined by the distance between two parts of the second spiral portion **29** that are adjacent to each other in the direction orthogonal to the spiraling direction. Preferably, the second winding pitch is equal to the first winding pitch of the first spiral portion **26**.

(71) The second spiral portion **29** can have any winding shape and the second inner region **67** can have any planar shape, which are thus not limited to those shown in FIG. **6** etc. The second spiral portion **29** can be wound in a polygonal shape, such as a triangular or quadrangular shape, or in a circular shape as seen in a plan view. The second inner region **67** can be defined, so as to fit the winding shape of the second spiral portion **29**, in a polygonal shape, such as a triangular or quadrangular shape, or in a circular shape as seen in a plan view.

(72) Preferably, the high-potential coil **23** is formed of the same conductive material as the low-potential coil **22**. That is, preferably, like the low-potential coil **22**, the high-potential coil **23** includes a barrier layer and a body layer.

(73) Referring to FIG. **4**, the semiconductor device **5** includes a plurality of (in the diagram, twelve) low-potential terminals **11** and a plurality of (in the diagram, twelve) high-potential terminals **12**. The plurality of low-potential terminals **11** are electrically connected to the low-potential coils **22** of the corresponding transformers **21A** to **21D** respectively. The plurality of high-potential terminals **12** are electrically connected to the high-potential coils **23** of the corresponding transformers **21A** to **21D** respectively.

(74) The plurality of low-potential terminals **11** are formed on the insulation principal surface **52** of the insulation layer **51**. Specifically, the plurality of low-potential terminals **11** are formed in a second insulation side wall **53B** side region, at an interval from the plurality of transformers **21A** to **21D** in the second direction Y, and are arrayed at intervals from each other in the first direction X.

(75) The plurality of low-potential terminals **11** include a first low-potential terminal **11A**, a second low-potential terminal **11B**, a third low-potential terminal **11C**, a fourth low-potential terminal **11D**, a fifth low-potential terminal **11E**, and a sixth low-potential terminal **11F**. Actually, in the embodiment, two each of the plurality of low-potential terminals **11A** to **11F** are formed. The plurality of low-potential terminals **11A** to **11F** may each include any number of terminals.

(76) The first low-potential terminal **11A** faces the first transformer **21A** in the second direction Y as seen in a plan view. The second low-potential terminal **11B** faces the second transformer **21B** in the second direction Y as seen in a plan view. The third low-potential terminal **11C** faces the third transformer **21C** in the second direction Y as seen in a plan view. The fourth low-potential terminal **11D** faces the fourth transformer **21D** in the second direction Y as seen in a plan view. The fifth low-potential terminal **11E** is formed in a region between the first and second low-potential terminals **11A** and **11B** as seen in a plan view. The sixth low-potential terminal **11F** is formed in a region between the third and fourth low-potential terminals **11C** and **11D** as seen in a plan view.

(77) The first low-potential terminal **11A** is electrically connected to the first inner end **24** of the first transformer **21A** (low-potential coil **22**). The second low-potential terminal **11B** is electrically connected to the first inner end **24** of the second transformer **21B** (low-potential coil **22**). The third low-potential terminal **11C** is electrically connected to the first inner end **24** of the third transformer **21C** (low-potential coil **22**). The fourth low-potential terminal **11D** is electrically connected to the first inner end **24** of the fourth transformer **21D** (low-potential coil **22**).

(78) The fifth low-potential terminal **11E** is electrically connected to the first outer end **25** of the first transformer **21A** (low-potential coil **22**) and to the first outer end **25** of the second transformer **21B** (low-potential coil **22**). The sixth low-potential terminal **11F** is electrically connected to the first outer end **25** of the third transformer **21C** (low-potential coil **22**) and to the first outer end **25** of the fourth transformer **21D** (low-potential coil **22**).

(79) The plurality of high-potential terminals **12** are formed on the insulation principal surface **52** of the insulation layer **51**, at an interval from the plurality of low-potential terminals **11**.

Specifically, the plurality of high-potential terminals **12** are formed in a first insulation side wall **53A** side region, at an interval from the plurality of low-potential terminals **11** in the second direction Y, and are arrayed at intervals from each other in the first direction X.

(80) The plurality of high-potential terminals **12** are formed in regions close to the corresponding transformers **21A** to **21D**, respectively, as seen in a plan view. The high-potential terminals **12** being close to the transformers **21A** to **21D** means that, as seen in a plan view, the distance between the high-potential terminals **12** and the transformers **21** is smaller than the distance between the low-potential terminals **11** and the high-potential terminals **12**.

(81) Specifically, as seen in a plan view, the plurality of high-potential terminals **12** are formed at intervals from each other along the first direction X so as to face the plurality of transformers **21A** to **21D** along the first direction X. More specifically, as seen in a plan view, the plurality of high-potential terminals **12** are formed at intervals from each other along the first direction X so as to be located in the second inner regions **67** in the high-potential coils **23** and in regions between adjacent high-potential coils **23**. As a result, as seen in a plan view, the plurality of high-potential terminals **12** are, along with the transformers **21A** to **21D**, arrayed in one row along the first direction X.

(82) The plurality of high-potential terminals **12** include a first high-potential terminal **12A**, a second high-potential terminal **12B**, a third high-potential terminal **12C**, a fourth high-potential terminal **12D**, a fifth high-potential terminal **12E**, and a sixth high-potential terminal **12F**. Actually, in the embodiment, two each of the plurality of high-potential terminals **12A** to **12F** are formed. The plurality of high-potential terminals **12A** to **12F** may each include any number of terminals.

(83) The first high-potential terminal **12A** is formed in the second inner region **67** in the first transformer **21A** (high-potential coil **23**) as seen in a plan view. The second high-potential terminal **12B** is formed in the second inner region **67** in the second transformer **21B** (high-potential coil **23**) as seen in a plan view. The third high-potential terminal **12C** is formed in the second inner region **67** in the third transformer **21C** (high-potential coil **23**) as seen in a plan view. The fourth high-potential terminal **12D** is formed in the second inner region **67** in the fourth transformer **21D** (high-potential coil **23**) as seen in a plan view. The fifth high-potential terminal **12E** is formed in a region between the first and second transformers **21A** and **21B** as seen in a plan view. The sixth high-potential terminal **12F** is formed in a region between the third and fourth transformers **21C** and **21D** as seen in a plan view.

(84) The first high-potential terminal **12A** is electrically connected to the second inner end **27** of the first transformer **21A** (high-potential coil **23**). The second high-potential terminal **12B** is electrically connected to the second inner end **27** of the second transformer **21B** (high-potential coil **23**). The third high-potential terminal **12C** is electrically connected to the second inner end **27** of the third transformer **21C** (high-potential coil **23**). The fourth high-potential terminal **12D** is electrically connected to the second inner end **27** of the fourth transformer **21D** (high-potential coil **23**).

(85) The fifth high-potential terminal **12E** is electrically connected to the second outer end **28** of the first transformer **21A** (high-potential coil **23**) and to the second outer end **28** of the second transformer **21B** (high-potential coil **23**). The sixth high-potential terminal **12F** is electrically connected to the second outer end **28** of the third transformer **21C** (high-potential coil **23**) and to the second outer end **28** of the fourth transformer **21D** (high-potential coil **23**).

(86) Referring to FIGS. 5 and 7, the semiconductor device **5** includes a first low-potential wiring **31**, a second low-potential wiring **32**, a first high-potential wiring **33**, and a second high-potential wiring **34**, all formed in the insulation layer **51**. Actually, in the embodiment, a plurality of first low-potential wirings **31**, a plurality of second low-potential wirings **32**, a plurality of first high-potential wirings **33**, and a plurality of second high-potential wirings **34** are formed.

(87) The first and second low-potential wirings **31** and **32** hold the low-potential coils **22** of the first and second transformers **21A** and **21B** at equal potentials. The first and second low-potential wirings **31** and **32** also hold the low-potential coils **22** of the third and fourth transformers **21C** and **21D** at equal potentials. In the embodiment, the first and second low-potential wirings **31** and **32** hold the low-potential coils **22** of all the transformers **21A** to **21D** at equal potentials.

(88) The first and second high-potential wirings **33** and **34** hold the high-potential coils **23** of the first and second transformers **21A** and **21B** at equal potentials. The first and second high-potential wirings **33** and **34** also hold the high-potential coils **23** of the third and fourth transformers **21C** and **21D** at equal potentials. In the embodiment, the first and second high-potential wirings **33** and **34** hold the high-potential coils **23** of all the transformers **21A** to **21D** at equal potentials.

(89) The plurality of first low-potential wirings **31** are electrically connected respectively to the corresponding low-potential terminals **11A** to **11D** and to the first inner ends **24** of the corresponding transformers **21A** to **21D** (low-potential coils **22**). The plurality of first low-potential wirings **31** have similar structures. In the following description, the structure of the first low-potential wiring **31** connected to the first low-potential terminal **11A** and to the first transformer **21A** will be described as an example. No separate description will be given of the structures of the other first low-potential wirings **31**, to which the description of the structure of the first low-potential wiring **31** connected to the first transformer **21A** is to be taken to apply.

(90) The first low-potential wiring **31** includes a through wiring **71**, a low-potential connection wiring **72**, a lead wiring **73**, a first connection plug electrode **74**, a second connection plug electrode **75**, one or a plurality of (in this embodiment, a plurality of) pad plug electrodes **76**, and one or a plurality of (in this embodiment, a plurality of) substrate plug electrodes **77**.

(91) Preferably, the through wiring **71**, the low-potential connection wiring **72**, the lead wiring **73**, the first connection plug electrode **74**, the second connection plug electrode **75**, the pad plug electrodes **76**, and the substrate plug electrodes **77** are formed of the same conductive material as the low-potential coil **22** and the like. That is, preferably, like the low-potential coil **22** and the like, the through wiring **71**, the low-potential connection wiring **72**, the lead wiring **73**, the first connection plug electrode **74**, the second connection plug electrode **75**, the pad plug electrodes **76**, and the substrate plug electrodes **77** each include a barrier layer and a body layer.

(92) The through wiring **71** penetrates a plurality of interlayer insulation layers **57** in the insulation layer **51** and extends in a columnar shape along the normal direction **Z**. In the embodiment, the through wiring **71** is formed in a region between the bottom and top insulation layers **55** and **56** in the insulation layer **51**. The through wiring **71** has a top end part at the top insulation layer **56** side and a bottom end part at the bottom insulation layer **55** side. The top end part of the through wiring **71** is formed in the same interlayer insulation layer **57** as the high-potential coil **23**, and is covered by the top insulation layer **56**. The bottom end part of the through wiring **71** is formed in the same interlayer insulation layer **57** as the low-potential coil **22**.

(93) In the embodiment, the through wiring **71** includes a first electrode layer **78**, a second electrode layer **79**, and a plurality of wiring plug electrodes **80**. In the through wiring **71**, the first and second electrode layers **78** and **79** and the wiring plug electrodes are formed of the same conductive material as the low-potential coil **22** and the like. That is, like the low-potential coil **22** and the like, the first and second electrode layers **78** and **79** and the wiring plug electrodes **80** each include a barrier layer and a body layer.

(94) The first electrode layer **78** constitutes the top end part of the through wiring **71**. The second electrode layer **79** constitutes the bottom end part of the through wiring **71**. The first electrode layer **78** is formed as an island, and faces the low-potential terminal **11** (first low-potential terminal **11A**) in the normal direction **Z**. The second electrode layer **79** is formed as an island, and faces the first electrode layer **78** in the normal direction **Z**.

(95) The plurality of wiring plug electrodes **80** are embedded respectively in the plurality of interlayer insulation layers **57** located in a region between the first and second electrode layers **78**

and **79**. The plurality of wiring plug electrodes **80** are stacked together from the bottom insulation layer **55** to the top insulation layer **56** so as to be electrically connected together, and electrically connect together the first and second electrode layers **78** and **79**. The plurality of wiring plug electrodes **80** each have a plane area smaller than the plane area of either of the first and second electrode layers **78** and **79**.

(96) The number of layers stacked in the plurality of wiring plug electrodes **80** is equal to the number of layers stacked in the plurality of interlayer insulation layer **57**. In the embodiment, six wiring plug electrodes **80** are embedded in interlayer insulation layers **57** respectively, and any number of wiring plug electrodes **80** can be embedded in interlayer insulation layers **57** respectively. Needless to say, one or a plurality of wiring plug electrodes **80** can be formed that penetrates a plurality of interlayer insulation layers **57**.

(97) The low-potential connection wiring **72** is formed in the same interlayer insulation layer **57** as the low-potential coil **22**, in the first inner region **66** in the first transformer **21A** (low-potential coil **22**). The low-potential connection wiring **72** is formed as an island, and faces the high-potential terminal **12** (first high-potential terminal **12A**) in the normal direction Z. Preferably, the low-potential connection wiring **72** has a plane area larger than the plane area of the wiring plug electrode **80**. The low-potential connection wiring **72** is electrically connected to the first inner end **24** of the low-potential coil **22**.

(98) The lead wiring **73** is formed in the interlayer insulation layer **57**, in a region between the semiconductor chip **41** and the through wiring **71**. In the embodiment, the lead wiring **73** is formed in the first interlayer insulation layer **57** as counted from the bottom insulation layer **55**. The lead wiring **73** has a first end part at one side, a second end part at the other side, and a wiring part that connects together the first and second end parts. The first end part of the lead wiring **73** is located in a region between the semiconductor chip **41** and the bottom end part of the through wiring **71**. The second end part of the lead wiring **73** is located in a region between the semiconductor chip **41** and the low-potential connection wiring **72**. The wiring part extends along the first principal surface **42** of the semiconductor chip **41**, and extends in the shape of a stripe in a region between the first and second end parts.

(99) The first connection plug electrode **74** is formed in the interlayer insulation layer **57**, in a region between the through wiring **71** and the lead wiring **73**, and is electrically connected to the through wiring **71** and to the first end part of the lead wiring **73**. The second connection plug electrode **75** is formed in the interlayer insulation layer **57**, in a region between the low-potential connection wiring **72** and the lead wiring **73**, and is electrically connected to the low-potential connection wiring **72** and to the second end part of the lead wiring **73**.

(100) The plurality of pad plug electrodes **76** are formed in the top insulation layer **56**, in a region between the low-potential terminal **11** (first low-potential terminal **11A**) and the through wiring **71**, and are electrically connected to the low-potential terminal **11** and to the top end part of the through wiring **71**. The plurality of substrate plug electrodes **77** are formed in the bottom insulation layer **55**, in a region between the semiconductor chip **41** and the lead wiring **73**. In the embodiment, the substrate plug electrodes **77** are formed in a region between the semiconductor chip **41** and the first end part of the lead wiring **73**, and are electrically connected to the semiconductor chip **41** and to the first end part of the lead wiring **73**.

(101) Referring to FIGS. **6** and **7**, the plurality of first high-potential wirings **33** are connected respectively to the corresponding high-potential terminals **12A** to **12D** and to the second inner ends **27** of the corresponding transformers **21A** to **21D** (high-potential coils **23**). The plurality of first high-potential wirings **33** have similar structures. In the following description, the structure of the first high-potential wiring **33** connected to the first high-potential terminal **12A** and to the first transformer **21A** will be described as an example. No description will be given of the structures of the other first high-potential wirings **33**, to which the description of the structure of the first high-potential wiring **33** connected to the first transformer **21A** is to be taken to apply.

(102) The first high-potential wiring **33** includes a high-potential connection wiring **81** and one or a plurality of (in this embodiment, a plurality of) pad plug electrodes **82**. Preferably, the high-potential connection wiring **81** and the pad plug electrodes **82** are formed of the same conductive material as the low-potential coil **22** and the like. That is, preferably, like the low-potential coil **22** and the like, the high-potential connection wiring **81** and the pad plug electrodes **82** each include a barrier layer and a body layer.

(103) The high-potential connection wiring **81** is formed in the same interlayer insulation layer **57** as the high-potential coil **23**, in the second inner region **67** in the high-potential coil **23**. The high-potential connection wiring **81** is formed as an island, and faces the high-potential terminal **12** (first high-potential terminal **12A**) in the normal direction Z. The high-potential connection wiring **81** is electrically connected to the second inner end **27** of the high-potential coil **23**. The high-potential connection wiring **81** is formed at an interval from the low-potential connection wiring **72** as seen in a plan view, and does not face the low-potential connection wiring **72** in the normal direction Z. This results in an increased insulation distance between the low- and high-potential connection wirings **72** and **81** and hence an increased dielectric strength voltage in the insulation layer **51**.

(104) The plurality of pad plug electrodes **82** are formed in the top insulation layer **56**, in a region between the high-potential terminal **12** (first high-potential terminal **12A**) and the high-potential connection wiring **81**, and are electrically connected to the high-potential terminal **12** and to the high-potential connection wiring **81**. The plurality of pad plug electrodes **82** each have a plane area smaller than the plane area of the high-potential connection wiring **81** as seen in a plan view.

(105) Referring to FIG. 7, preferably, the distance **D1** between the low- and high-potential terminals **11** and **12** is larger than the distance **D2** between the low- and high-potential coils **22** and **23** ($D2 < D1$). Preferably, the distance **D1** is larger than the total thickness **DT** of the plurality of interlayer insulation layers **57** ($DT < D1$). The ratio $D2/D1$ of the distance **D2** to the distance **D1** can be 0.01 or more but 0.1 or less. Preferably, the distance **D1** is 100 μm or more but 500 μm or less. The distance **D2** can be 1 μm or more but 50 μm or less. Preferably, the distance **D2** is 5 μm or more but 25 μm or less. The distances **D1** and **D2** can have any values, which are adjusted appropriately according to the desired dielectric strength voltage.

(106) Referring to FIGS. 6 and 7, the semiconductor device **5** has a dummy pattern **85** that is embedded in the insulation layer **51** so as to be located around the transformers **21A** to **21D** as seen in a plan view.

(107) The dummy pattern **85** is formed in a pattern different (discontinuous) from that of either of the high- and low-potential coils **23** and **22**, and is independent of the transformers **21A** to **21D**. That is, the dummy pattern **85** does not function as part of the transformers **21A** to **21D**. The dummy pattern **85** is formed as a shield conductor layer that shields electric fields between the low- and high-potential coils **22** and **23** in the transformers **21A** to **21D** to suppress electric field concentration on the high-potential coil **23**. In the embodiment, the dummy pattern **85** is patterned at a line density per unit area that is equal to the line density of the high-potential coil **23**. The line density of the dummy pattern **85** being equal to the line density of the high-potential coil **23** means that the line density of the dummy pattern **85** falls within the range of $\pm 20\%$ of the line density of the high-potential coil **23**.

(108) The dummy pattern **85** can be formed at any depth in the insulation layer **51**, which is adjusted according to the electric field strength to be attenuated. Preferably, the dummy pattern **85** is formed in a region closer to the high-potential coil **23** than to the low-potential coil **22** with respect to the normal direction Z. The dummy pattern **85** being closer to the high-potential coil **23** with respect to the normal direction Z means that, with respect to the normal direction Z, the distance between the dummy pattern **85** and the high-potential coil **23** is smaller than the distance between the dummy pattern **85** and the low-potential coil **22**.

(109) In that way, electric field concentration on the high-potential coil **23** can be suppressed properly. The smaller the distance between the dummy pattern **85** and the high-potential coil **23**

with respect to the normal direction Z, the more effectively electric field concentration on the high-potential coil **23** can be suppressed. Preferably, the dummy pattern **85** is formed in the same interlayer insulation layer **57** as the high-potential coil **23**. In that way, electric field concentration on the high-potential coil **23** can be suppressed more properly. The dummy pattern **85** includes a plurality of dummy patterns that are in varying electrical states. The dummy pattern **85** can include a high-potential dummy pattern.

(110) The high-potential dummy pattern **86** can be formed at any depth in the insulation layer **51**, which is adjusted according to the electric field strength to be attenuated. Preferably, the high-potential dummy pattern **86** is formed in a region closer to the high-potential coil **23** than to the low-potential coil **22** with respect to the normal direction Z. The high-potential dummy pattern **86** being closer to the high-potential coil **23** with respect to the normal direction Z means that, with respect to the normal direction Z, the distance between the high-potential dummy pattern **86** and the high-potential coil **23** is smaller than the distance between the high-potential dummy pattern **86** and the low-potential coil **22**.

(111) The dummy pattern **85** includes a floating dummy pattern that is formed in an electrically floating state in the insulation layer **51** so as to be located around the transformers **21A** to **21D**.

(112) In the embodiment, the floating dummy pattern is patterned in dense lines so as to partly cover and partly expose a region around the high-potential coil **23** as seen in a plan view. The floating dummy pattern can be formed so as to have ends or no ends.

(113) The floating dummy pattern can be formed at any depth in the insulation layer **51**, which is adjusted according to the electric field strength to be attenuated.

(114) Any number of floating lines can be provided, which is adjusted according to the electric field strength to be attenuated. The floating dummy pattern can include a plurality of floating lines.

(115) Referring to FIG. 7, the semiconductor device **5** includes a second functional device **60** that is formed in the first principal surface **42** of the semiconductor chip **41** in a device region **62**. The second functional device **60** is formed using a superficial part of the first principal surface **42** and/or a region on the first principal surface **42** of the semiconductor chip **41**, and is covered by the insulation layer **51** (bottom insulation layer **55**). In FIG. 7, the second functional device **60** is shown in a simplified form by broken lines indicated in a superficial part of the first principal surface **42**.

(116) The second functional device **60** is electrically connected to a low-potential terminal **11** via a low-potential wiring, and is electrically connected to a high-potential terminal **12** via a high-potential wiring. Except that the low-potential wiring is patterned in the insulation layer **51** so as to be connected to the second functional device **60**, it has a similar structure to the first low-potential wiring **31** (second low-potential wiring **32**). Except that the high-potential wiring is patterned in the insulation layer **51** so as to be connected to the second functional device **60**, it has a similar structure to the first high-potential wiring **33** (second high-potential wiring **34**). No description will be given of the low- and high-potential wirings associated with the second functional device **60**.

(117) The second functional device **60** can include at least one of a passive device, a semiconductor rectification device, and a semiconductor switching device. The second functional device **60** can include a circuit network comprising a selective combination of any two or more of a passive device, a semiconductor rectification device, and a semiconductor switching device. The circuit network can constitute part or the whole of an integrated circuit.

(118) The passive device can include a semiconductor passive device. The passive device can include one or both of a resistor and a capacitor. The semiconductor rectification device can include at least one of a pn-junction diode, a PIN diode, a Zener diode, a Schottky barrier diode, and a fast-recovery diode. The semiconductor switching device can include at least one of a BJT (bipolar junction transistor), a MISFET (metal-insulator-semiconductor field-effect transistor), an IGBT (insulated-gate bipolar junction transistor), and a JFET (junction field-effect transistor).

(119) Referring to FIGS. 5 to 7, the semiconductor device **5** further includes a sealing conductor **61**

embedded in the insulation layer **51**. The sealing conductor **61** is embedded in the form of walls in the insulation layer **51**, at intervals from the insulation side walls **53A** to **53D** as seen in a plan view, and partitions the insulation layer **51** into the device region **62** and an outer region **63**. The sealing conductor **61** prevents moisture entry and crack development from the outer region **63** to the device region **62**.

(120) The device region **62** is a region that includes the first functional device **45** (plurality of transformers **21**), the second functional device **60**, the plurality of low-potential terminals **11**, the plurality of high-potential terminals **12**, the first low-potential wirings **31**, the second low-potential wirings **32**, the first high-potential wirings **33**, the second high-potential wirings **34**, and the dummy pattern **85**. The outer region **63** is a region outside the device region **62**.

(121) The sealing conductor **61** is electrically isolated from the device region **62**. Specifically, the sealing conductor **61** is electrically isolated from the first functional device **45** (plurality of transformers **21**), the second functional device **60**, the plurality of low-potential terminals **11**, the plurality of high-potential terminals **12**, the first low-potential wirings **31**, the second low-potential wirings **32**, the first high-potential wirings **33**, the second high-potential wirings **34**, and the dummy pattern **85**. More specifically, the sealing conductor **61** is held in an electrically floating state. The sealing conductor **61** does not form a current path connected to the device region **62**.

(122) The sealing conductor **61** is formed in the shape of a stripe along the insulation side walls **53A** to **53D**. In the embodiment, the sealing conductor **61** is formed in a quadrangular ring shape (specifically, a rectangular ring shape) as seen in a plan view. Thus, the sealing conductor **61** defines the outer region **63** in a quadrangular ring shape (specifically, a rectangular ring shape) surrounding the device region **62** as seen in a plan view.

(123) Specifically, the sealing conductor **61** has a top end part at the insulation principal surface **52** side, a bottom end part at the semiconductor chip **41** side, and a wall part that extends in the form of walls between the top and bottom end parts. In the embodiment, the top end part of the sealing conductor **61** is formed at an interval from the insulation principal surface **52** toward the semiconductor chip **41**, and is located in the insulation layer **51**. In the embodiment, the top end part of the sealing conductor **61** is covered by the top insulation layer **56**. The top end part of the sealing conductor **61** can be covered by one or a plurality of interlayer insulation layers **57**. The top end part of the sealing conductor **61** can be exposed through the top insulation layer **56**. The bottom end part of the sealing conductor **61** is formed at an interval from the semiconductor chip **41** toward the top end part.

(124) Thus, in the embodiment, the sealing conductor **61** is embedded in the insulation layer **51** so as to be located at the semiconductor chip **41** side of the plurality of low-potential terminals **11** and the plurality of high-potential terminals **12**. Moreover, in the insulation layer **51**, the sealing conductor **61** faces, in the direction parallel to the insulation principal surface **52**, the first functional device **45** (plurality of transformers **21**), the first low-potential wirings **31**, the second low-potential wirings **32**, the first high-potential wirings **33**, the second high-potential wirings **34**, and the dummy pattern **85**. In the insulation layer **51**, the sealing conductor **61** can face, in the direction parallel to the insulation principal surface **52**, part of the second functional device **60**.

(125) The sealing conductor **61** includes a plurality of sealing plug conductors **64** and one or a plurality of (in the embodiment, a plurality of) sealing via conductors **65**. Any number of sealing via conductors **65** may be provided. Of the plurality of sealing plug conductors **64**, the top sealing plug conductor **64** constitutes the top end part of the sealing conductor **61**. The plurality of sealing via conductors **65** constitute the bottom end part of the sealing conductor **61**. Preferably, the sealing plug conductors **64** and the sealing via conductors **65** are formed of the same conductive material as the low-potential coil **22**. That is, preferably, like the low-potential coil **22** and the like, the sealing plug conductors **64** and the sealing via conductors **65** each include a barrier layer and a body layer.

(126) The plurality of sealing plug conductors **64** are embedded in the plurality of interlayer

insulation layers **57** respectively, and are each formed in a quadrangular ring shape (specifically, a rectangular ring shape) surrounding the device region **62**. The plurality of sealing plug conductors **64** are stacked together from the bottom insulation layer **55** to the top insulation layer **56** so as to be connected together. The number of layers stacked in the plurality of sealing plug conductors **64** is equal to the number of layers in the plurality of interlayer insulation layers **57**. Needless to say, one or a plurality of sealing plug conductors **64** may be formed that penetrates a plurality of interlayer insulation layers **57**.

(127) So long as a set of a plurality of sealing plug conductor **64** constitutes one ring-shaped sealing conductor **61**, not all the sealing plug conductors **64** need be formed in a ring shape. For example, at least one of the plurality of sealing plug conductors **64** can be formed so as to have ends. Or at least one of the plurality of sealing plug conductors **64** may be divided into a plurality of strip-shaped portions with ends. However, with consideration given to the risk of moisture entry and crack development into the device region **62**, preferably, the plurality of sealing plug conductors **64** are formed so as to have no ends (in a ring shape).

(128) The plurality of sealing via conductors **65** are formed in the bottom insulation layer **55**, in a region between the semiconductor chip **41** and the sealing plug conductors **64**. The plurality of sealing via conductors **65** are formed at an interval from the semiconductor chip **41**, and are connected to the sealing plug conductors **64**. The plurality of sealing via conductors **65** have a plane area smaller than the plane area of the sealing plug conductors **64**. In a case where a single sealing via conductor **65** is formed, the single sealing via conductors **65** can have a plane area larger than the plane area of the sealing plug conductors **64**.

(129) The sealing conductor **61** can have a width of 0.1 μm or more but 10 μm or less. Preferably, the sealing conductor **61** has a width of 1 μm or more but 5 μm or less. The width of the sealing conductor **61** is defined by its width in the direction orthogonal to the direction in which it extends.

(130) Referring to FIGS. 7 and 8, the semiconductor device **5** further includes a separation structure **130** that is interposed between the semiconductor chip **41** and the sealing conductor **61** and that electrically isolates the sealing conductor **61** from the semiconductor chip **41**. Preferably, the separation structure **130** includes an insulator. In the embodiment, the separation structure **130** is a field insulation film **131** formed on the first principal surface **42** of the semiconductor chip **41**.

(131) The field insulation film **131** includes at least one of an oxide film (silicon oxide film) and a nitride film (silicon nitride film). Preferably, the field insulation film **131** is a LOCOS (local oxidation of silicon) film as one example of an oxide film that is formed through oxidation of the first principal surface **42** of the semiconductor chip **41**. The field insulation film **131** can have any thickness so long as it can insulate between the semiconductor chip **41** and the sealing conductor **61**. The field insulation film **131** can have a thickness of 0.1 μm or more but 5 μm or less.

(132) The separation structure **130** is formed on the first principal surface **42** of the semiconductor chip **41**, and extends in the shape of a stripe along the sealing conductor **61** as seen in a plan view. In the embodiment, the separation structure **130** is formed in a quadrangular ring shape (specifically, a rectangular ring shape) as seen in a plan view. The separation structure **130** has a connection portion **132** to which the bottom end part of the sealing conductor **61** (i.e., the sealing via conductors **65**) is connected. The connection portion **132** can form an anchor portion into which the bottom end part of the sealing conductor **61** (i.e., the sealing via conductors **65**) is anchored toward the semiconductor chip **41**. Needless to say, the connection portion **132** can be formed to be flush with the principal surface of the separation structure **130**.

(133) The separation structure **130** includes an inner end part **130A** at the device region **62** side, an outer end part **130B** at the outer region **63** side, and a main body part **130C** between the inner and outer end parts **130A** and **130B**. As seen in a plan view, the inner end part **130A** defines the region where the second functional device **60** is formed (i.e., the device region **62**). The inner end part **130A** can be formed integrally with an insulation film (not illustrated) formed on the first principal surface **42** of the semiconductor chip **41**.

(134) The outer end part **130B** is exposed on the chip side walls **44A** to **44D** of the semiconductor chip **41**, and is continuous with the chip side walls **44A** to **44D** of the semiconductor chip **41**. More specifically, the outer end part **130B** is formed so as to be flush with the chip side walls **44A** to **44D** of the semiconductor chip **41**. The outer end part **130B** constitutes a polished surface between, to be flush with, the chip side walls **44A** to **44D** of the semiconductor chip **41** and the insulation side walls **53A** to **53D** of the insulation layer **51**. Needless to say, an embodiment is also possible where the outer end part **130B** is formed within the first principal surface **42** at intervals from the chip side walls **44A** to **44D**.

(135) The main body part **130C** has a flat surface that extends substantially parallel to the first principal surface **42** of the semiconductor chip **41**. The main body part **130C** has the connection portion **132** to which the bottom end part of the sealing conductor **61** (i.e., the sealing via conductors **65**) is connected. The connection portion **132** is formed in the main body part **130C**, at intervals from the inner and outer end parts **130A** and **130B**. The separation structure **130** can be implemented in many ways other than in the form of a field insulation film **131**.

(136) Referring to FIG. 7, the semiconductor device **5** further includes an inorganic insulation layer **140** formed on the insulation principal surface **52** of the insulation layer **51** so as to cover the sealing conductor **61**. The inorganic insulation layer **140** can be called a passivation layer. The inorganic insulation layer **140** protects the insulation layer **51** and the semiconductor chip **41** from above the insulation principal surface **52**.

(137) In the embodiment, the inorganic insulation layer **140** has a stacked structure composed of a first inorganic insulation layer **141** and a second inorganic insulation layer **142**. The first inorganic insulation layer **141** can contain silicon oxide. Preferably, the first inorganic insulation layer **141** contains USG (undoped silicate glass), which is undoped silicon oxide. The first inorganic insulation layer **141** can have a thickness of 50 nm or more but 5000 nm or less. The second inorganic insulation layer **142** can contain silicon nitride. The second inorganic insulation layer **142** can have a thickness of 500 nm or more but 5000 nm or less. Increasing the total thickness of the inorganic insulation layer **140** helps increase the dielectric strength voltage above the high-potential coils **23**.

(138) In a configuration where the first inorganic insulation layer **141** is made of USG and the second inorganic insulation layer **142** is made of silicon nitride, USG has the higher dielectric breakdown voltage (V/cm) than silicon nitride. In view of this, when thickening the inorganic insulation layer **140**, it is preferable to form the first inorganic insulation layer **141** thicker than the second inorganic insulation layer **142**.

(139) The first inorganic insulation layer **141** can contain at least one of BPSG (boron-doped phosphor silicate glass) and PSG (phosphorus silicate glass) as examples of silicon oxide. In that case, however, since the silicon oxide contains a dopant (boron or phosphorus), for an increased dielectric strength voltage above the high-potential coils **23**, it is particularly preferable to form the first inorganic insulation layer **141** of USG. Needless to say, the inorganic insulation layer **140** can have a single-layer structure composed of either the first or second inorganic insulation layer **141** or **142**.

(140) The inorganic insulation layer **140** covers the entire area of the sealing conductor **61**, and has a plurality of low-potential pad openings **143** and a plurality of high-potential pad openings **144** that are formed in a region outside the sealing conductor **61**. The plurality of low-potential pad openings **143** expose the plurality of low-potential terminals **11** respectively. The plurality of high-potential pad openings **144** expose the plurality of high-potential terminals **12** respectively. The inorganic insulation layer **140** can have overlap parts that overlap circumferential edge parts of the low-potential terminals **11**. The inorganic insulation layer **140** can have overlap parts that overlap circumferential edge parts of the high-potential terminals **12**.

(141) The semiconductor device **5** further includes an organic insulation layer **145** that is formed on the inorganic insulation layer **140**. The organic insulation layer **145** can contain photosensitive

resin. The organic insulation layer **145** can contain at least one of polyimide, polyamide, and polybenzoxazole. In the embodiment, the organic insulation layer **145** contains polyimide. The organic insulation layer **145** can have a thickness of 1 μm or more but 50 μm or less.

(142) Preferably, the organic insulation layer **145** has a thickness larger than the total thickness of the inorganic insulation layer **140**. Moreover, preferably, the inorganic and organic insulation layers **140** and **145** together have a total thickness larger than the distance D2 between the low- and high-potential coils **22** and **23**. In that case, preferably, the inorganic insulation layer **140** has a total thickness of 2 μm or more but 10 μm or less. Preferably, the organic insulation layer **145** has a thickness of 5 μm or more but 50 μm or less. Such structures help suppress an increase in the thicknesses of the inorganic and organic insulation layers **140** and **145** while appropriately increasing the dielectric strength voltage above the high-potential coil **23** owing to the stacked film of the inorganic and organic insulation layers **140** and **145**.

(143) The organic insulation layer **145** includes a first part **146** that covers a low-potential side region and a second part **147** that covers a high-potential side region. The first part **146** covers the sealing conductor **61** across the inorganic insulation layer **140**. The first part **146** has a plurality of low-potential terminal openings **148** through which the plurality of low-potential terminals **11** (low-potential pad openings **143**) are respectively exposed in a region outside the sealing conductor **61**. The first part **146** can have overlapping parts that overlap circumferential edges (overlap parts) of the low-potential pad openings **143**.

(144) The second part **147** is formed at an interval from the first part **146**, and exposes the inorganic insulation layer **140** between the first and second parts **146** and **147**. The second part **147** has a plurality of high-potential terminal openings **149** through which the plurality of high-potential terminals **12** (high-potential pad openings **144**) are respectively exposed. The second part **147** can have overlap parts that overlap circumferential edges (overlap parts) of the high-potential pad openings **144**.

(145) The second part **147** covers the transformers **21A** to **21D** and the dummy pattern **85** together. Specifically, the second part **147** covers the plurality of high-potential coils **23**, the plurality of high-potential terminals **12**, a first high-potential dummy pattern **87**, a second high-potential dummy pattern **88**, and a floating dummy pattern **121** together.

(146) The present invention can be implemented in any other embodiments. The embodiment described above deals with an example where a first functional device **45** and a second functional device **60** are formed. An embodiment is however also possible that only has a second functional device **60**, with no first functional device **45**. In that case, the dummy pattern **85** may be omitted. This structure provides, with respect to the second functional device **60**, effects similar to those mentioned in connection with the first embodiment (except those associated with the dummy pattern **85**).

(147) That is, in a case where a voltage is applied to the second functional device **60** via the low- and high-potential terminals **11** and **12**, it is possible suppress unnecessary conduction between the high-potential terminal **12** and the sealing conductor **61**. Likewise, in a case where a voltage is applied to the second functional device **60** via the low- and high-potential terminals **11** and **12**, it is possible suppress unnecessary conduction between the low-potential terminal **11** and the sealing conductor **61**.

(148) The embodiment described above deals with an example where a second functional device **60** is formed. The second functional device **60** however is not essential, and can be omitted.

(149) The embodiment described above deals with an example where a dummy pattern **85** is formed. The dummy pattern **85** however is not essential, and can be omitted.

(150) The embodiment described above deals with an example where the first functional device **45** is of a multichannel type that includes a plurality of transformers **21**. It is however also possible to employ a single-channel first functional device **45** that includes a single transformer **21**.

Transformer Layout

(151) FIG. 9 is a plan view (top view) schematically showing one example of transformer layout in a two-channel transformer chip **300** (corresponding to the semiconductor device **5** described previously). The transformer chip **300** shown there includes a first transformer **301**, a second transformer **302**, a third transformer **303**, a fourth transformer **304**, a first guard ring **305**, a second guard ring **306**, pads **a1** to **a8**, pads **b1** to **b8**, pads **c1** to **c4**, and pads **d1** to **d4**.

(152) In the transformer chip **300**, the pads **a1** and **b1** are connected to one terminal of the secondary coil **L1s** of the first transformer **301**, and the pads **c1** and **d1** are connected to the other terminal of that secondary coil **L1s**. The pads **a2** and **b2** are connected to one terminal of the secondary coil **L2s** of the second transformer **302**, and the pads **c1** and **d1** are connected to the other terminal of that secondary coil **L2s**.

(153) Moreover, the pads **a3** and **b3** are connected to one terminal of the secondary coil **L3s** of the third transformer **303**, and the pads **c2** and **d2** are connected to the other terminal of that secondary coil **L3s**. The pads **a4** and **b4** are connected to one terminal of the secondary coil **L4s** of the fourth transformer **304**, and the pads **c2** and **d2** are connected to the other terminal of that secondary coil **L4s**.

(154) FIG. 9 does not show any of the primary coils of the first, second, third, and fourth transformers **301**, **302**, **303**, and **304**. The primary coils basically have structures similar to those of the secondary coils **L1s** to **L4s** respectively, and are disposed right below the secondary coils **L1s** to **L4s**, respectively, so as to face them.

(155) Specifically, the pads **a5** and **b5** are connected to one terminal of the primary coil of the first transformer **301**, and the pads **c3** and **d3** are connected to the other terminal of that primary coil. Likewise, the pads **a6** and **b6** are connected to one terminal of the primary coil of the second transformer **302**, and the pads **c3** and **d3** are connected to the other terminal of that primary coil.

(156) Likewise, the pads **a7** and **b7** are connected to one terminal of the primary coil of the third transformer **303**, and the pads **c4** and **d4** are connected to the other terminal of that primary coil. Likewise, the pads **a8** and **b8** are connected to one terminal of the primary coil of the fourth transformer **304**, and the pads **c4** and **d4** are connected to the other terminal of that primary coil.

(157) The pads **a5** to **a8**, the pads **b5** to **b8**, the pads **c3** and **c4**, and the pads **d3** and **d4** mentioned above are each led from inside the transformer chip **300** to its surface across an unillustrated via.

(158) Of the plurality of pads mentioned above, the pads **a1** to **a8** each correspond to a first current feed pad, and the pads **b1** to **b8** each correspond to a first voltage measurement pad; the pads **c1** to **c4** each correspond to a second current feed pad, and the pads **d1** to **d4** each correspond to a second voltage measurement pad.

(159) Thus, the transformer chip **300** of this configuration example permits, during its defect inspection, accurate measurement of the series resistance component across each coil. It is thus possible not only to reject defective products with a broken wire in a coil but also to appropriately reject defective products with an abnormal resistance value in a coil (e.g., a midway short circuit between coils), and hence to prevent defective products from being distributed in the market.

(160) For a transformer chip **300** that has passed the defect inspection mentioned above, the plurality of pads described above can be used for connection with a primary-side chip and a secondary-side chip (e.g., the controller chip **210** and the driver chip **220** described previously).

(161) Specifically, the pads **a1** and **b1**, the pads **a2** and **b2**, the pads **a3** and **b3**, and the pads **a4** and **b4** can each be connected to one of the signal input and output terminals of the secondary-side chip; the pads **c1** and **d1** and the pads **c2** and **d2** can each be connected to a common voltage application terminal (**GND2**) of the secondary-side chip.

(162) On the other hand, the pads **a5** and **b5**, the pads **a6** and **b6**, the pads **a7** and **b7**, and the pads **a8** and **b8** can each be connected to one of the signal input and output terminals of the primary-side chip; the pads **c3** and **d3** and the pads **c4** and **d4** can each be connected to a common voltage application terminal (**GND1**) of the primary-side chip.

(163) Here, as shown in FIG. 9, the first to fourth transformers **301** to **304** are so arranged as to be

coupled for each signal transmission direction. In terms of what is shown in the diagram, for example, the first and second transformers **301** and **302**, which transmit a signal from the primary-side chip to the secondary-side chip, are coupled into a first pair by the first guard ring **305**. Likewise, for example, the third and fourth transformers **303** and **302**, which transmit a signal from the secondary-side chip to the primary-side chip, are coupled into a second pair by the second guard ring **306**.

(164) Such coupling is intended, in a structure where the primary and secondary coils of each of the first to fourth transformers **301** to **304** are formed so as to be stacked on each other in the up-down direction of the substrate, to obtain a desired withstand voltage between the primary and secondary coils. The first and second guard rings **305** and **306** are however not essential elements.

(165) The first and second guard rings **305** and **306** can be connected via pads e1 and e2, respectively, to a low-impedance wiring such as a grounded terminal.

(166) In the transformer chip **300**, the pads c1 and d1 are shared between the secondary coils L1s and L2s. The pads c2 and d2 are shared between the secondary coils L3s and L4s. The pads c3 and d3 are shared between the primary coils L1p and L2p. The pads c4 and d4 are shared between the primary coils that correspond to them respectively. This configuration helps reduce the number of pads and helps make the transformer chip **300** compact.

(167) Moreover, as shown in FIG. 9, the primary and secondary coils of the first to fourth transformers **301** to **304** are preferably each wound in a rectangular shape (or, with the corners rounded, in a running-track shape) as seen in a plan view of the transformer chip **300**. This configuration helps increase the area over which the primary and secondary coils overlap each other and helps enhance the transmission efficiency across the transformers.

(168) Needless to say, the illustrated transformer layout is merely an example; any number of coils of any shape can be disposed in any layout, and pads can be disposed in any layout. Any of the chip structure, transformer layouts, etc. described above can be applied to semiconductor devices in general that have a coil integrated in a semiconductor chip.

Traction Inverter

(169) FIG. 10 is a diagram showing the basic configuration of a traction inverter incorporated in an electric vehicle. The traction inverter **400** of this configuration example is a kind of motor driving device that converts direct-current electric power supplied from an unillustrated vehicle-mounted battery into alternating-current electric power to drive a motor M. The traction inverter **400** includes an isolated gate driver **1**, an ECU (electronic control unit) **2**, and various discrete components (a high-side switch SWH, a low-side switch SWL, an npn-type bipolar transistor Q1, a pnp-type bipolar transistor Q2, resistors R1 to R3, and capacitors C1 and C2).

(170) Examples of electric vehicles (what is called xEVs) in which the traction inverter **400** can be incorporated include BEVs (battery electric vehicles), HEVs (hybrid electric vehicles), PHEVs (plug-in hybrid electric vehicles)/PHVs (plug-in hybrid vehicles), and FCEVs (fuel cell electric vehicles)/FCVs (fuel cell vehicles).

(171) The isolated gate driver **1** is a semiconductor integrated circuit device (e.g., corresponding to the signal transmission device **200** in FIG. 1) that, while isolating between a primary circuit system (VCC1-GND1) supplied with electric power from a direct-current voltage source E1 and a secondary circuit system (VCC2-GND2) that is supplied with electric power from a direct-current voltage source E2, transmits a gate driving signal from the primary circuit system to the secondary circuit system.

(172) The isolated gate driver **1** has, as means for establishing electrical connection with outside it, a plurality of external terminals (of which some are shown in the diagram, namely a VCC1 terminal, an IN terminal, an FLT1 terminal, an FLT2 terminal, a GND1 terminal, a VCC2 terminal, an OUT terminal, a CLAMP terminal, a PROOUT terminal, and a GND2 terminal).

(173) The VCC1 terminal is the power terminal of the primary circuit system. The IN terminal is a control input terminal. The FLT1 and FLT2 terminals are each a fault signal output terminal. The

GND1 terminal is the ground terminal of the primary circuit system. The VCC2 terminal is the power terminal of the secondary circuit system. The OUT terminal is an output terminal. The CLAMP terminal is a mirror clamp terminal. The PROOUT terminal is a soft turn-off terminal. The GND2 terminal is the ground terminal of the secondary circuit system.

(174) The internal configuration and the operation of the isolated gate driver **1** will be described in detail later.

(175) The ECU **2** serves as a means for comprehensive electrical control of an electric vehicle, and engages in exchange of various signals (an input signal IN, external fault signals FLT1 and FLT2, etc.) with the isolated gate driver **1**.

(176) The discrete components are interconnected as follows. The resistor R1 is connected between the VCC1 terminal and the FLT1 terminal. The resistor R2 is connected between the VCC1 terminal and the FLT2 terminal. The resistor R3 is connected between the CLAMP terminal and the PROOUT terminal. The capacitor C1 is connected between the VCC1 terminal and the GND1 terminal. The capacitor C2 is connected between the VCC2 terminal and the GND2 terminal.

(177) The collector of the transistor Q1 is connected to the VCC2 terminal. The emitters of the transistors Q1 and Q2 and the control terminal (e.g., gate) of the high-side switch SWH are all connected to the CLAMP terminal. The collector of the transistor Q2 is connected to the GND2 terminal. The bases of the transistors Q1 and Q2 are both connected to the OUT terminal.

(178) The high-side and low-side switches SWH and SWL are connected, respectively, between an application terminal for a first motor driving voltage VD1 and an input terminal of the motor M corresponding to the relevant phase (specifically, one end of the motor coil of that phase) and between the input terminal of the motor M corresponding to the relevant phase and an application terminal for a second motor driving voltage VD2 (<VD1). As these switches SWH and SWL are turned on and off, they control the supply of a motor driving current, thus constituting a half-bridge output stage.

(179) In the traction inverter **400** of this configuration example, the high-side and low-side switches SWH and SWL are implemented with insulated-gate bipolar transistors (IGBTs). This, however, is not meant to limit the configuration of the half-bridge output stage: instead, they may be implemented with MOS (metal-oxide-semiconductor) field-effect transistors using a SiC (silicon carbide) semiconductor or MOS field-effect transistors using a Si semiconductor. In particular, MOS field-effect transistors using a SiC, for their low power consumption and high heat resistance as compared with MOS field-effect transistors using a Si semiconductor, are suitable for use on electric vehicles.

(180) For the sake of convenience, the diagram only shows the half-bridge output stage corresponding to one phase and the isolated gate driver **1** connected to the high-side switch SWH. In reality, an isolated gate driver **1** is connected to each of the high-side and low-side switches SWH and SWL constituting the half-bridge output stage of each phase. Accordingly, in a case where the motor M is a three-phase AC (alternating-current) motor, there are provided half-bridge output stages for three phases, and these require a total of six isolated gate drivers **1**.

Isolated Gate Driver (Basic Configuration)

(181) With reference still to FIG. **10**. The isolated gate driver **1** will be described in detail. The isolated gate driver **1** has a first semiconductor chip **410**, a second semiconductor chip **420**, and a third semiconductor chip **430** sealed in a single package.

(182) The first semiconductor chip **410** is a controller chip that has integrated in it a controller that is driven by being supplied with a supply voltage VCC1 (e.g., 5 V or 3.3 V relative to GND1) from the direct-current voltage source E1 to generate switch control signals S1 and S2 based on an input signal IN. The first semiconductor chip **410** has the following main functions: generating the switch control signals S1 and S2; generating external fault signals FLT1 and FLT2; and performing UVLO (undervoltage lock-out). The first semiconductor chip **410** can be designed to have an appropriate withstand voltage (e.g., 7 [V]) with consideration given to the supply voltage VCC1

(relative to GND1).

(183) The second semiconductor chip **420** is a driver chip that has integrated in it a driver that is driven by being supplied with a supply voltage VCC2 (10 to 30 V relative to GND2) from the direct-current voltage source E2 to control the driving of the high-side switch SWH, of which one terminal is fed with a voltage as high as several hundred volts, based on the switch control signals S1 and S2 fed from the first semiconductor chip **410** via the third semiconductor chip **430**. The second semiconductor chip **420** has the following main functions: generating an output signal OUT; generating internal fault signals S3 and S4; and performing UVLO. The second semiconductor chip **420** can be designed to have an appropriate withstand voltage (e.g., 40 [V]) with consideration given to the supply voltage VCC2 (relative to GND2).

(184) The third semiconductor chip **430** is a transformer chip that has integrated in it a transformer that, while isolating between the first and second semiconductor chips **410** and **420** on a direct-current basis, exchanges the switch control signals S1 and S2 and the internal fault signals S3 and S4.

(185) As described above, the isolated gate driver **1** of this configuration example has, separately from the first semiconductor chip **410** having a controller integrated in it and the second semiconductor chip **420** having a driver integrated in it, the transformer chip **430** incorporating a transformer alone, and these three chips are sealed in a single package.

(186) With this configuration, the first and second semiconductor chips **410** and **420** can each be formed by a common low-withstand-voltage process (with a withstand voltage of several volts to several tens of volts). This eliminates the need for a dedicated high-withstand-voltage process (with a withstand voltage of several kilovolts), and helps reduce manufacturing costs.

(187) Moreover, the first and second semiconductor chips **410** and **420** can each be fabricated by a time-proven existing process. This eliminates the need for conducting reliability tests anew, and contributes to a shortened development period and reduced development costs.

(188) Next, the internal configuration of each of the first, second, and third semiconductor chips **410**, **420**, and **430** incorporated in the isolated gate driver **1** will be described in detail one by one.

(189) The first semiconductor chip **410** includes a first transmitter **411**, a second transmitter **412**, a first receiver **413**, a second receiver **414**, a logic circuit **415**, a first UVLO circuit **416**, and N-channel MOS field-effect transistors Na and Nb.

(190) The second semiconductor chip **420** includes a third receiver **421**, a fourth receiver **422**, a third transmitter **423**, a fourth transmitter **424**, a logic circuit **425**, a driver **426**, a second UVLO circuit **427**, a P-channel MOS field-effect transistor P1, N-channel MOS field-effect transistors N1 to N3, and an SR flip-flop FF.

(191) The third semiconductor chip **430** includes a first transformer **431**, a second transformer **432**, a third transformer **433**, and a fourth transformer **434**.

(192) The first transmitter **411** transmits the switch control signal S1, which is fed from the logic circuit **415**, to the third receiver **421** via the first transformer **431**.

(193) The second transmitter **412** transmits the switch control signal S2, which is fed from the logic circuit **415**, to the fourth receiver **422** via the second transformer **432**.

(194) The first receiver **413** receives the internal fault signal S3, which is fed from the third transmitter **423** via the third transformer **433**, to deliver the internal fault signal S3 to the logic circuit **415**.

(195) The second receiver **414** receives the internal fault signal S4, which is fed from the fourth transmitter **424** via the fourth transformer **434**, to deliver the internal fault signal S4 to the logic circuit **415**.

(196) The logic circuit **415** exchanges various signals (the input signal IN and the external fault signals FLT1 and FLT2) with the ECU **2**, and exchanges various signals (S1 to S4) with the second semiconductor chip **420** by using the first transmitter **411**, the second transmitter **412**, the first receiver **413**, and the second receiver **414**.

(197) If the input signal IN is at high level, the logic circuit **415** generates a pulse in the switch control signal S1 to turn the output signal OUT to high level; by contrast, if the input signal IN is at low level, the logic circuit **415** generates a pulse in the switch control signal S2 to turn the output signal OUT to low level. For example, the logic circuit **415** raises a pulse in the switch control signal S1 by detecting a positive edge (a rising edge from low level to high level) in the input signal IN, and raises a pulse in the switch control signal S2 by detecting a negative edge (a falling edge from high level to low level) in the input signal IN.

(198) Moreover, according to the state of the isolated gate driver **1**, the logic circuit **415** turns the transistors Na and Nb on and off to switch the logic levels of the external fault signals FLT1 and FLT2.

(199) The drains of the transistors Na and Nb are connected to the FLT1 and FLT2 terminals respectively. The sources of the transistors Na and Nb are both connected to the ground terminal (i.e., the GND1 terminal) of the primary circuit system. The gates of the transistors Na and Nb are both connected to the logic circuit **415**.

(200) For example, so long as the isolated gate driver **1** has no fault, the transistors Na and Nb are both off, so that the FLT1 and FLT2 are both in a high-impedance state (a state pulled up by the resistor R1). In this state, the external fault signals FLT1 and FLT2 are both at high level (=VCC1). By contrast, if the isolated gate driver **1** has some fault, at least one of the transistors Na and Nb is on, so that at least one of the FLT1 and FLT2 is in a state short-circuited to the ground. In this state, at least one of the external fault signals FLT1 and FLT2 is at low level (=GND1). With this configuration, the ECU **2** can, by monitoring the external fault signals FLT1 and FLT2, grasp the state of the isolated gate driver **1**. This function, fault output function, will be described in detail later.

(201) The first UVLO circuit **416** monitors whether the supply voltage VCC1 is in an undervoltage state and delivers the monitoring result to the logic circuit **415**.

(202) The third receiver **421** receives the switch control signal S1, which is fed from the first transmitter **411** via the first transformer **431**, to deliver the switch control signal S1 to the set input terminal (S) of the SR flip-flop FF.

(203) The fourth receiver **422** receives the switch control signal S2, which is fed from the second transmitter **412** via the second transformer **432**, to deliver the switch control signal S2 to the reset input terminal (R) of the SR flip-flop FF.

(204) The third transmitter **423** transmits the internal fault signal S3, which is fed from the logic circuit **425**, to the first receiver **413** via the third transformer **433**.

(205) The fourth transmitter **424** transmits the internal fault signal S4, which is fed from the logic circuit **425**, to the second receiver **414** via the fourth transformer **434**.

(206) Triggered by a pulse edge in the switch control signal S1 fed to its set input terminal (S), the SR flip-flop FF sets the logic level at its output terminal (Q) to high level. Triggered by a pulse edge in the switch control signal S2 fed to its reset input terminal (R), the SR flip-flop FF resets the logic level at its output terminal (Q) to low level. Thus, the output signal fed from the SR flip-flop FF to the logic circuit **425** is a pulse signal that has the same logic level as the input signal IN fed from the ECU **2** to the logic circuit **415**.

(207) Based on the output signal of the SR flip-flop FF, the logic circuit **425** generates a driving signal for the driver **426**. If the second UVLO circuit **427** detects an undervoltage fault, the logic circuit **425** sends a corresponding notification directly to the driver **426** and also, by using the internal fault signal S3, to the logic circuit **415**. With this configuration, even if the second semiconductor chip **420** has an undervoltage fault, the driver **426** can promptly perform protection operation, and the logic circuit **415** can perform fault output to the ECU **2**.

(208) Though not specifically shown in the diagram, the second semiconductor chip **420** may include any fault detector other than the second UVLO circuit **427** (such as an overvoltage detector OVP [overvoltage protection], a short-circuit detector SCP [short-circuit protection], or an

overtemperature detector OTP [overtemperature protection]). In that case, if any fault is detected in the second semiconductor chip **420**, as in the case described above, protection operation can be performed for the driver **426** and fault output can be performed to the logic circuit **415** (hence the ECU **2**) by use of the internal fault signal **S3**.

(209) As will be described in detail later, the second semiconductor chip **420** may include a non-volatile memory for use as registers. In that case, if an error is detected in the data stored in the non-volatile memory, fault output may be performed to the logic circuit **415** (hence the ECU **2**) by use of, for example, the internal fault signal **S4** provided separately from the internal fault signal **S3** mentioned above.

(210) The source of the transistor **P1** is connected to the **VCC2** terminal. The drains of the transistors **P1** and **N1** are both connected to the **OUT** terminal. The drain of the transistor **N2** is connected to the **CLAMP** terminal. The drain of the transistor **N3** is connected to the **PROOUT** terminal. The sources of the transistors **N1** to **N3** are all connected to the **GND2** terminal. The gates of the transistors **P1** and **N1** to **N3** are all connected to the driver **426**.

(211) The driver **426** turns on and off the transistors **P1** and **N1** based on a driving signal fed from the logic circuit **425** to output the output signal **OUT** from the connection node between the transistors **P1** and **N1**. The output signal **OUT** is fed via a driving circuit constituted by the transistors **Q1** and **Q2** to the high-side switch **SWH**. In the driving circuit just mentioned, the rising/falling time (slew rate) of the output signal **OUT** is set such that the output signal **OUT** has the capacity to drive the high-side switch **SWH**. Here, when the output signal **OUT** is at high level, the high-side switch **SWH** is on; when the output signal **OUT** is at low level, the high-side switch **SWH** is off.

(212) The driver **426** has a function (active mirror clamp function) of, in response to the voltage level of the output signal **OUT** (relative to **GND2**) turning to low level, turning on the transistor **N2** so as to draw electric charge (mirror current) from the gate of the high-side switch **SWH** via the **CLAMP** terminal. With this configuration, when the high-side switch **SWH** is turned off, it is possible to promptly drop the gate potential of the high-side switch **SWH** to low level via the transistor **N2** regardless of the slew rate set by the driving circuit mentioned above.

(213) The driver **426** also has a function (soft turn-on function) of, on recognizing the need for protection operation based on a fault detection signal fed from the logic circuit **425**, turning the transistors **P1**, **N1**, and **N2** all off and the transistor **N3** on. Such switching permits electric charge to be drawn more gently from the gate of the high-side switch **SWH** via the resistor **R3** during protection operation than during normal operation. This configuration helps avoid an instantaneous break of the motor current during protection operation, and thus helps suppress a surge ascribable to a back electromotive force in the motor coil. The falling time during protection operation can be adjusted as desired by appropriately selecting the resistance value of the resistor **R3**.

(214) The second UVLO circuit **427** monitors whether the supply voltage **VCC2** is in an undervoltage state and delivers the monitoring result to the logic circuit **425**.

(215) The first transformer **431** is a direct-current isolation element for transmitting the switch control signal **S1** from the first semiconductor chip **410** to the second semiconductor chip **420**.

(216) The second transformer **432** is a direct-current isolation element for transmitting the switch control signal **S2** from the first semiconductor chip **410** to the second semiconductor chip **420**.

(217) The third transformer **433** is a direct-current isolation element for transmitting the internal fault signal **S3** from the second semiconductor chip **420** to the first semiconductor chip **410**.

(218) The fourth transformer **434** is a direct-current isolation element for transmitting the internal fault signal **S4** from the second semiconductor chip **420** to the first semiconductor chip **410**.

(219) With this configuration where not only the switch control signals **S1** and **S2** but also the internal fault signals **S3** and **S4** are exchanged between the first and second semiconductor chips **410** and **420**, it is possible not only to turn on and off the high-side switch **SWH** but also to carry out various fault detection functions appropriately.

(220) Here is a review of the different functions of the isolated gate driver **1** of this configuration example.

UVLO1 (Controller-Side Undervoltage-Induced Malfunctioning Prevention Function)

(221) If the supply voltage V_{CC1} becomes equal to or lower than a predetermined low-side threshold voltage $V_{sub.UVLO1L}$, the isolated gate driver **1** turns off the high-side switch SWH and keeps the FLT1 terminal at low level. If the supply voltage V_{CC1} becomes equal to or higher than a predetermined high-side threshold voltage $V_{sub.UVLO1H}$, the isolated gate driver **1** starts normal operation and keeps the FLT1 terminal in a high-impedance state (at high level).

UVLO2 (Driver-Side Undervoltage-Induced Malfunctioning Prevention Function)

(222) If the supply voltage V_{CC2} becomes equal to or lower than a predetermined low-side threshold voltage $V_{sub.UVLO2L}$, the isolated gate driver **1** turns off the high-side switch SWH and keeps the FLT1 terminal at low level. If the supply voltage V_{CC2} becomes equal to or higher than a predetermined high-side threshold voltage $V_{sub.UVLO2H}$, the isolated gate driver **1** starts normal operation and keeps the FLT1 terminal in a high-impedance state (at high level).

Soft Turn-Off During Protection Operation

(223) When forcibly turning off the high-side switch SWH, the isolated gate driver **1** turns the PROOUT terminal to low level and brings the OUT terminal into a high-impedance state. This control permits the high-side switch SWH to be turned off slowly. The slew rate during its turning-off can be adjusted as desired by appropriately selecting the resistance value of the externally connected resistor R3.

Active Mirror Clamp

(224) When the gate potential of the high-side switch SWH becomes equal to or lower than a predetermined threshold voltage $V_{sub.AMC}$, the isolated gate driver **1** turns the CLAMP terminal to low level. This control permits reliable turning-off of the high-side switch SWH.

A Study on Set Size and Cost Reduction

(225) In the traction inverter **400**, optimizing the switching characteristics of a switching element (e.g., the high-side switch SWH) implemented with an IGBT or the like and the operating threshold values of a fault detector (such as an overvoltage detector OVP, short-circuit detector SCP, or overtemperature detector OTP) will lead to reducing safety margins against heating and destruction of the switching element. This helps reduce the chip size of the switching element and thus helps achieve size reduction and cost reduction of sets.

(226) Specifically, a characteristic of the isolated gate driver **1**, for example the driving speed of the gate driver (i.e., the slew rate), or the detection threshold value or cancellation threshold value of a fault detector, can be left adjustable as desired according to the individual variation (manufacturing variation) of the switching element. Presented below will be a novel embodiment devised to achieve that.

Isolated Gate Driver (Embodiment)

(227) FIG. **11** is a diagram showing a novel embodiment of the isolated gate driver **1**. The isolated gate driver **1** according to this embodiment includes a gate driver **441**, a fault detector-controller **442**, a register **443**, a control logic **444**, a non-volatile memory **445**, a memory controller **446**, and an interface **447**. The following description assumes basically that these components are integrated in the second semiconductor chip **420** (driver chip) in the isolated gate driver **1**.

(228) The gate driver **441** is a circuit block that corresponds to the driver **426** in FIG. **10**. The gate driver **441** generates the output signal OUT based on the driving signal fed from the logic circuit **425** (not shown in the diagram), thereby to drive the gate of the high-side switch SWH. The gate driver **441** is switched between an enabled and a disabled state based on a fault detection signal fed from the fault detector-controller **442**. The characteristics of the gate driver **441** can be adjusted as desired based on values (in the diagram, adjustment data D1) stored in the register **443**. The characteristics include, for example, the driving speed of the gate driver **441**, that is, the rising/falling time (slew rate) of the output signal OUT.

(229) The fault detector-controller **442** has combined functions as a fault detector (such as an overvoltage detector OVP, short-circuit detector SCP, or overtemperature detector OTP) that detects a fault elsewhere than in the non-volatile memory **445** and as a fault controller that feeds the external fault signals FLT1 and FLT2 to the ECU **2** via the FLT1 and FLT2 terminals. For example, the FLT1 terminal corresponds to a first external terminal for feeding out the result of fault detection elsewhere than in the non-volatile memory **445**, and the FLT2 terminal corresponds to a second external terminal for feeding out the result of error detection in the non-volatile memory **445**. The characteristics of the fault detector-controller **442** can be adjusted as desired according to values (in the diagram, adjustment data D2) stored in the register **443**. The characteristics include, for example, the detection threshold value or cancellation threshold value referred to in a check for a fault.

(230) Here, the fault detector-controller **442** includes, as its components, a fault signal transmission mechanism for transmitting fault detection results from the second semiconductor chip **420** to the first semiconductor chip **410** (e.g., the first receiver **413**, the second receiver **414**, the third transmitter **423**, the fourth transformer **434**, the third transformer **433**, and the fourth transformer **434** in FIG. **10**) and a fault terminal control mechanism for controlling the FLT1 and FLT2 terminals (e.g., the logic circuit **415** and the transistors Na and Nb in FIG. **10**). Accordingly, the fault detector-controller **442** can be understood not to have all its components integrated in the second semiconductor chip **420** but have its components integrated in, in a fashion distributed among, the first, second, and third semiconductor chips **410**, **420**, and **430**.

(231) The register **443** stores on a non-volatile basis the adjustment data read from the non-volatile memory **445** by the control logic **444** (i.e., in the diagram, the adjustment data D1 for the gate driver **441** and the adjustment data D2 for the fault detector-controller **442**).

(232) Every time the traction inverter **400** starts up (i.e., every time the isolated gate driver **1** starts receiving power), the control logic **444** reads the adjustment data D1 and D2 from the non-volatile memory **445** and stores it in the register **443**. Here, the control logic **444** has a function of controlling the fault detector-controller **442** so as to keep the gate driver **441** in a disabled state until the adjustment data D1 and D2 read from the non-volatile memory **445** completes being stored in the register **443** (this will be described in detail later). Also, the control logic **444** incorporates an error detection-correction circuit ECC (error check and correct) that performs error detection and error correction with respect to the adjustment data D1 and D2 written to the non-volatile memory **445** (this will be described in detail later).

(233) The non-volatile memory **445** stores on a non-volatile basis the adjustment data for the isolated gate driver **1** (i.e., in the diagram, the adjustment data D1 for the gate driver **441** and the adjustment data D2 for the fault detector-controller **442**). While in this embodiment the non-volatile memory **445** is incorporated in the isolated gate driver **1**, it may instead be externally connected to the isolated gate driver **1** (a specific example will be described later).

(234) The memory controller **446** performs address control and the like with respect to the non-volatile memory **445** when the non-volatile memory **445** is accessed (e.g., to read or write the adjustment data D1 and D2) from outside the isolated gate driver **1** via the interface **447**.

(235) The interface **447** is a front end via which to access the non-volatile memory **445** from outside the isolated gate driver **1**. The interface **447** can be implemented with, for example, a two-line serial interface complying with the I2C (inter-integrated circuit) standard that conducts bidirectional serial communication by use of a data signal SDA and a clock signal SCL.

(236) With the isolated gate driver **1** according to this embodiment, by reading the adjustment data D1 and D2 previously written to the non-volatile memory **445** and storing it in the register **443**, it is possible to optimize, based on the values stored in the register **443**, the characteristics of each of the gate driver **441** and the fault detector-controller **442**. It is thus possible to reduce margins against heating and destruction of the switching element (e.g. the high-side switch SWH), and thus to reduce the chip size of the switching element and achieve size reduction and cost reduction of

sets.

(237) If, however, the gate driver **441** operates before the characteristics of each of the gate driver **441** and the fault detector-controller **442** complete being set, that is, in other words, before the adjustment data **D1** and **D2** completes being stored in the register **443**, the switching element may be driven with inappropriate characteristics with no consideration given to the individual (manufacturing) variation of the switching element. In such a situation, the reduced safety margins can act adversely, leading to failure of the switching element.

(238) Considering in particular that the register **443** is a kind of volatile memory, the above sequence of operation of reading the adjustment data **D1** and **D2** from the non-volatile memory **445** and storing it in the register **443** is repeated every time the traction inverter **400** is powered up. Thus, the start-up sequence from powering-up to the start of gate driving needs to be well designed.

Start-Up Sequence

(239) FIG. **12** is a flow chart showing the start-up sequence of the isolated gate driver **1**. The sequence proceeds as follows. At step #1, a set (i.e., the traction inverter **400**) is powered up, so that the isolated gate driver **1** starts to be supplied with the supply voltages **VCC1** and **VCC2**.

(240) At step #2, as the supply voltages **VCC1** and **VCC2** rise, the ULVO on the isolated gate driver **1** is cancelled. At this point, however, the reading of the adjustment data **D1** and **D2** from the non-volatile memory **445** has not been started yet and thus, naturally, its storing in the register **443** and hence the adjustment of the characteristics of each of the gate driver **441** and the fault detector-controller **442** has not been completed. Accordingly, the gate driver **441** is kept in a disabled state. The gate driver **441** can be kept in a disabled state, for example, by keeping the fault detection signal fed from the fault detector-controller **442** to the gate driver **441** at a logic level indicating a fault being detected.

(241) At step #3, the control logic **444** reads the adjustment data **D1** and **D2** from the non-volatile memory **445** and stores it in the register **443**.

(242) At step #4, a check is made for completion of the storing of data in the register **443** and hence completion of the adjustment of the characteristics of each of the gate driver **441** and the fault detector-controller **442**. If the check result is “yes”, the sequence proceeds to step #5; if the check result is “no”, the sequence returns to step #3 to continue, with the gate driver **441** kept in a disabled state, the storing of data in the register **443**.

(243) If, at step #3, the storing of data in the register **443** is complete, the characteristics of each of the gate driver **441** and the fault detector-controller **442** are optimized and thus the switching element (e.g., the high-side switch **SWH**) is now ready to be driven. Accordingly, if the sequence advances via step #4 resulting in “yes” to step #5, the isolated gate driver **1** becomes fault-released and the fault detection signal fed to the gate driver **441** is turned from the logic level indicating a fault being detected to a logic level indicating no fault being detected.

(244) As a result, at the subsequent step, #6, the gate driver **441** is released from the disabled state and the switching element starts to be driven and controlled according to the input signal **IN** (normal operation).

(245) As will be understood from the sequence of operation described above, the isolated gate driver **1** employs a start-up sequence that keeps the gate driver **441** in a disabled state until the characteristics of each of the gate driver **441** and the fault detector-controller **442** complete being set, in other words, until the adjustment data **D1** and **D2** completes being stored in the register **443**.

(246) With this configuration, even if the input signal **IN** is fed in from the ECU **2** before the setting of the characteristics of each of the gate driver **441** and the fault detector-controller **442**, the switching element is prevented from being driven with inappropriate characteristics with no consideration given to its individual (manufacturing) variation. Thus, operating the switching element with reduced safety margins is less likely to cause failure of the switching element.

Examples of Implementation of a Non-Volatile Memory

(247) FIG. **13** is a diagram showing a first implementation example of the non-volatile memory

445. The traction inverter **400** of the first implementation example is, as mentioned earlier, a kind of motor driving device that controls the driving of a motor M, and includes three isolated gate drivers **1H**(u/v/w), three isolated gate drivers **1L**(u/v/w), three high-side switches SWH(u/v/w), three low-side switches SWL(u/v/w), and an ECU **2**.

(248) The motor M is a three-phase motor that is driven to rotate according to three-phase driving voltages U/V/W fed from half-bridge output stages corresponding to three phases (U phase/V phase/W phase).

(249) The isolated gate drivers **1H**(u/v/w), while isolating between the ECU **2** and the high-side switches SWH(u/v/w) respectively, generate high-side gate drive signals (corresponding to the output pulse signal OUT mentioned above) according to the high-side gate control signals (corresponding to the input signal IN mentioned above) fed from the ECU **2**, and thereby drive the high-side switches SWH(u/v/w).

(250) The isolated gate drivers **1L**(u/v/w), while isolating between the ECU **2** and the low-side switches SWL(u/v/w) respectively, generate low-side gate drive signals according to the low-side gate control signals fed from the ECU **2**, and thereby drive the low-side switches SWL(u/v/w).

(251) The high-side switches SWH(u/v/w) are high-side power transistors that constitute half-bridge output stages of three phases (U phase/V phase/W phase) respectively, and are externally connected to the switch connection terminals T1 of the corresponding isolated gate drivers **1H**(u/v/w) (i.e., the OUT terminal, CLAMP terminal, PROOUT terminal, GND2 terminal, etc. provided at the second semiconductor chip **420** side in FIG. **10**) so that their gates are driven by the isolated gate drivers **1H**(u/v/w) respectively. The high-side switches SWH(u/v/w) are connected between the power-system power supply terminal (an application terminal for the first motor driving voltage VD1) and the input terminals of the motor M of the corresponding phases.

(252) The low-side switches SWL(u/v/w) are low-side power transistors that constitute half-bridge output stages of three phases (U phase/V phase/W phase) respectively, and are externally connected to the switch connection terminals T1 of the corresponding isolated gate drivers **1L**(u/v/w) (i.e., the OUT terminal, CLAMP terminal, PROOUT terminal, GND2 terminal, etc. provided at the second semiconductor chip **420** side in FIG. **10**) so that their gates are driven by the isolated gate drivers **1L**(u/v/w) respectively. The low-side switches SWL(u/v/w) are connected between the input terminals of the motor M of the corresponding phases and the power-system ground terminal (an application terminal for the second motor driving voltage VD2).

(253) While in the diagram the high-side and low-side switches SWH(u/v/w) and SWL(u/v/w) are implemented with IGBTs, as mentioned earlier they may instead be implemented with SiC-MOSFETs or Si-MOSFETs.

(254) The ECU **2** drives via the isolated gate drivers **1H**(u/v/w) and **1L**(u/v/w) the high-side and low-side switches SWH(u/v/w) and SWL(u/v/w) respectively, and thereby controls the driving of the rotation of the motor M. The ECU **2** also has a function of monitoring the FLT1 and FLT2 terminals of each of the isolated gate drivers **1H**(u/v/w) and **1L**(u/v/w) to achieve various kinds of safety control.

(255) Here, the isolated gate driver **1** described previously can be used suitably as any of the isolated gate drivers **1H**(u/v/w) and **1L**(u/v/w).

(256) In the traction inverter **400** of the first implementation example, the isolated gate drivers **1H**(u/v/w) and **1L**(u/v/w) each incorporate a non-volatile memory **445** for storing the corresponding adjustment data D1 and D2. This configuration helps reduce the number of components of the traction inverter **400**.

(257) FIG. **14** is a diagram showing a second implementation example of the non-volatile memory **445**. The traction inverter **400** of the second implementation example is based on the first implementation example (FIG. **13**), a difference being that the non-volatile memory **445** is externally connected.

(258) In terms of what is shown in the diagram, specifically, in the traction inverter **400** of the

second implementation example, to memory connection terminals T2 provided respectively at the secondary side (at the second semiconductor chip **420** side in FIG. **10**) of the isolated gate drivers **1H(u/v/w)** and **1L(u/v/w)**, non-volatile memories **445** each having the corresponding adjustment data **D1** and **D2** written to it are externally connected, on a one-memory-to-one-driver basis. With this configuration, the adjustment data **D1** and **D2** written to the non-volatile memory **445** on the secondary side can be used to adjust the characteristics of each of the gate driver **441** and the fault detector-controller **442**, which are provided on the same, secondary, side. This eliminates the need for signal transmission from the primary side to the secondary side, that is, signal transmission from the first semiconductor chip **410** to the second semiconductor chip **420** via the third semiconductor chip **430**. This helps simplify the circuit configuration of the isolated gate driver **1**. (259) FIG. **15** is a diagram showing a third implementation example of the non-volatile memory **445**. The traction inverter **400** of the third implementation example is based on the first implementation example (FIG. **13**), a difference being that the non-volatile memory **445** is externally connected.

(260) In terms of what is shown in the diagram, specifically, in the traction inverter **400** of the third implementation example, to memory connection terminals T2 provided respectively at the primary side (at the first semiconductor chip **410** side in FIG. **10**) of the isolated gate drivers **1H(u/v/w)** and **1L(u/v/w)**, a single non-volatile memory **445** having the corresponding adjustment data **D1** and **D2** written to it is externally connected, on a one-memory-to-all-drivers basis. With this configuration, it is possible to reduce the number of non-volatile memories **445** that are externally connected. It should however be noted that this configuration still requires signal transmission from the primary side to the secondary side, that is, signal transmission from the first semiconductor chip **410** to the second semiconductor chip **420** via the third semiconductor chip **430**.

A Study on Deterioration of a Non-Volatile Memory

(261) As the non-volatile memory **445** deteriorates with time, the adjustment data **D1** and **D2** written to it may be lost. When this occurs, not all of a plurality of bits are lost at once but they are lost bit by bit. To stay prepared for that, it is preferable that the control logic **444** (or non-volatile memory **445**) be provided with an error detection-correction circuit ECC that performs error detection and error correction with respect to the adjustment data **D1** and **D2** written to the non-volatile memory **445**.

(262) With the just-mentioned error detection-correction circuit ECC incorporated, if a one-bit error occurs in the adjustment data **D1** and **D2** written to the non-volatile memory **445**, it can be detected and corrected. It is thus possible to properly perform characteristics adjustment for the isolated gate driver **1** and continue the traveling of the electric vehicle without any problem. However, if a two-or-more-bit error occurs in the adjustment data **D1** and **D2** written to the non-volatile memory **445**, the error detection-correction circuit ECC can no longer correct them. This makes it impossible to properly perform characteristics adjustment for the isolated gate driver **1** and continue safe traveling of the electric vehicle.

(263) In view of the above study, it is preferable to switch the fault output from the isolated gate driver **1** according to whether the adjustment data **D1** and **D2** has a one-bit error or a two-or-more-bit error. Presented below will be examples of operation for appropriate fault output.

Fault Output

(264) FIG. **16** is a table showing a first operation example of fault output, showing the error detection result with respect to the non-volatile memory **445**, the enabled/disabled state of the gate driver **441**, and the output states at the FLT1 and FLT2 terminals respectively.

(265) Error detection with respect to the non-volatile memory **445** is carried out chiefly by the error detection-correction circuit ECC. Enabled/disabled state switching for the gate driver **441** and fault output via the FLT1 and FLT2 terminals are carried out chiefly by the fault detector-controller **442**.

(266) As shown at line **1**, when the adjustment data **D1** and **D2** written to the non-volatile memory **445** is normal, the gate driver **441** performs normal operation. In this case, the FLT1 and FLT2

terminals are both in a high-impedance state, and thus the external fault signals FLT1 and FLT2 delivered to the ECU 2 are, by being pulled-up by the resistors R1 and R2, at high level (i.e., the logic level indicating normal condition).

(267) As shown at line 2, if a one-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445, the FLT2 terminal is kept at low level (the logic level indicating an error being detected) and the gate driver 441 continues normal operation. The FLT1 terminal remains in a high-impedance state. In this way, a one-bit error can be detected and corrected by the error detection-correction circuit ECC, and thus the normal operation of the gate driver 441, and hence the traveling of the electric vehicle, can be continued with no problem.

(268) Even if a one-bit error is detected in the adjustment data D1 and D2, characteristics adjustment for the isolated gate driver 1 can be performed properly; thus fault output for the ECU 2 (i.e., output of a low-level external fault signal FLT2) does not necessarily have to be performed. Even so, by performing such fault output in anticipation of an upcoming error in another bit, it is possible to previously notify the electric vehicle's driver of the likelihood of a breakdown. It is thus possible to prevent the electric vehicle from breaking down suddenly, with no previous sign, on occurrence of a two-bit error.

(269) As shown at line 3, if a two-or-more-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445, the FLT1 and FLT2 terminals are both kept at low level and the gate driver 441 is forcibly stopped. In this way, if a two-or-more-bit error occurs, it is impossible to correct the adjustment data D1 and D2 and thus to perform characteristics adjustment for the isolated gate driver 1 properly. It is therefore of the highest priority to forcibly stop the gate driver 441 to ensure the safety of the electric vehicle.

(270) Moreover, by dropping the external fault signal FLT1 to low level, it is possible to promptly notify the ECU 2 of the occurrence of some serious fault in the isolated gate driver 1 or the high-side switch SWH (here, a two-or-more-bit error in the non-volatile memory 445 and the resulting forcible stop of the gate driver 441).

(271) FIG. 17 is a table showing a second operation example of fault output and, like FIG. 16 referred to previously, shows the error detection result with respect to the non-volatile memory 445, the enabled/disabled state of the gate driver 441, and the output states at the FLT1 and FLT2 terminals respectively.

(272) The second operation example shown in the diagram is basically similar to the first operation example (FIG. 16) described previously, the only difference being that, if a two-or-more-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445, the FLT2 terminal is kept in a high-impedance state. In this way, on detection of a two-or-more-bit error, the FLT1 terminal is used to perform fault output for the ECU 2; thus, the output state at the FLT2 terminal does not matter.

(273) FIG. 18 is a table showing a third operation example of fault output, and shows the error detection result with respect to the non-volatile memory 445, the fault detection result with respect to other than the non-volatile memory 445 (such as UVLO, OVP, SCP, and OTP), the enabled/disabled state of the gate driver 441, and the output states at the FLT1 and FLT2 terminals respectively.

(274) Error detection with respect to the non-volatile memory 445 is carried out chiefly by the error detection-correction circuit ECC. On the other hand, fault detection with respect to other than the non-volatile memory 445 (such as UVLO, OVP, SCP, and OTP), enabled/disabled state switching for the gate driver 441, and fault output via the FLT1 and FLT2 terminals are carried out chiefly by the fault detector-controller 442.

(275) As shown at line 1, if the adjustment data D1 and D2 written to the non-volatile memory 445 is normal and in addition no fault is detected with respect to other than the non-volatile memory 445, the gate driver 441 performs normal operation. In this case, the FLT1 and FLT2 terminals are both in a high-impedance state, and thus the external fault signals FLT1 and FLT2 delivered to the

ECU 2 are, by being pulled-up by the resistors R1 and R2, at high level (the logic level indicating normal condition).

(276) As shown at line 2, if the adjustment data D1 and D2 written to the non-volatile memory 445 is normal but a fault is detected with respect to other than the non-volatile memory 445, then, regardless of the error detection result, the FLT1 terminal is kept at low level (the output state indicating a fault being detected) and the gate driver 441 is forcibly stopped. It is thus possible, while ensuring the safety of the electric vehicle, to promptly notify the ECU 2 of the occurrence of some serious fault (such as UVLO, OVP, SCP, or OTP) in the isolated gate driver 1 or the high-side switch SWH. Since the adjustment data D1 and D2 has no error, the FLT2 terminal remains in a high-impedance state.

(277) As shown at line 3, if a one-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445 and in addition no fault is detected elsewhere than in the non-volatile memory 445, the FLT2 terminal is kept at low level (the logic level indicating an error being detected) and the gate driver 441 continues normal operation. The FLT1 terminal remains in a high-impedance state. In this way, a one-bit error can be detected and corrected by the error detection-correction circuit ECC, and thus the normal operation of the gate driver 441, and hence the traveling of the electric vehicle, can be continued with no problem. Moreover, with the FLT2 terminal dropped to low level, it is possible to notify the electric vehicle's driver of a sign of a breakdown.

(278) As shown at line 4, if a one-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445 and in addition also a fault is detected elsewhere than in the non-volatile memory 445, the FLT1 and FLT2 terminals are both kept at low level and the gate driver 441 is forcibly stopped. In this way, if a fault is detected elsewhere than in the non-volatile memory 445, as in the case shown at line 1 and described previously, regardless of the error detection result, the FLT1 terminal is kept at low level (the output state indicating a fault being detected), the gate driver 441 is forcibly stopped. It is thus possible, while ensuring the safety of the electric vehicle, to promptly notify the ECU 2 of the occurrence of some serious fault (such as UVLO, OVP, SCP, or OTP) in the isolated gate driver 1 or the high-side switches SWH.

(279) As shown at line 5, if a two-or-more-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445, then, regardless of the fault detection result with respect to other than the non-volatile memory 445, the FLT1 and FLT2 terminals are both kept at low level and the gate driver 441 is forcibly stopped. In this way, if a two-or-more-bit error occurs, it is impossible to correct the adjustment data D1 and D2 and thus to perform characteristics adjustment for the isolated gate driver 1 properly. It is therefore of the highest priority to forcibly stop the gate driver 441 to ensure the safety of the electric vehicle.

(280) Moreover, by dropping the external fault signal FLT1 to low level, it is possible to promptly indicate the ECU 2 of the occurrence of some serious fault in the isolated gate driver 1 or the high-side switch SWH (here, a two-or-more-bit error in the non-volatile memory 445 and the resulting forcible stop of the gate driver 441). In these respects, this operation example is no different from the first and second operation examples (FIGS. 16 and 17 respectively) described previously.

(281) FIG. 19 is a table showing a fourth operation example of fault output and, like FIG. 18 referred to previously, shows the error detection result with respect to the non-volatile memory 445, the fault detection result with respect to other than the non-volatile memory 445 (such as UVLO, OVP, SCP, and OTP), the enabled/disabled state of the gate driver 441, and the output states at the FLT1 and FLT2 terminals respectively.

(282) The fourth operation example shown in the diagram is basically similar to the third operation example (FIG. 18) described previously, the only difference being that, if a two-or-more-bit error is detected in the adjustment data D1 and D2 written to the non-volatile memory 445, the FLT2 terminal is kept in a high-impedance state. In this way, on detection of a two-or-more-bit error, the FLT1 terminal is used to perform fault output for the ECU 2; thus, the output state at the FLT2

terminal does not matter.

Sharing of a Fault Signal Output Terminal

(283) In the first to fourth operation examples (FIGS. **16** to **19**) described previously, if with respect to the adjustment data **D1** and **D2** written to the non-volatile memory **445** a one-bit error is detected, the **FLT2** terminal is used to perform fault output (sign-of-breakdown notification) and, if a two-or-more-bit error is detected, the **FLT1** terminal is used to perform fault output (breakdown notification). Here, these two kinds of fault output can be performed via the **FLT2** terminal alone, while the **FLT1** terminal is left uninvolved. Now, with reference to the relevant diagrams, a description will be given of examples in which the **FLT2** terminal is shared.

(284) FIG. **20** is a diagram showing a first example of sharing a fault signal output terminal (here, the **FLT2** terminal). As shown there, the fault detector-controller **442** can, if the adjustment data **D1** and **D2** is normal, keep the **FLT2** terminal at high level (e.g., **VCC1**); if a one-bit error is detected in the adjustment data **D1** and **D2**, keep the **FLT2** terminal at middle level (e.g., **VCC1/2**); and if a two-or-more-bit error is detected in the adjustment data **D1** and **D2**, keep the **FLT2** terminal at low level (e.g., **GND1**). With this configuration, it is possible to recognize the result of error detection based on the voltage level at the **FLT2** terminal.

(285) FIG. **21** is a diagram showing a second example of sharing a fault signal output terminal (here, the **FLT2** terminal). As shown there, the fault detector-controller **442** can, if the adjustment data **D1** and **D2** is normal, hold the **FLT2** terminal at high level; if a one-bit error is detected in the adjustment data **D1** and **D2**, pulse-drive (oscillate) the **FLT2** terminal between high and low levels; and if a two-or-more-bit error is detected in the adjustment data **D1** and **D2**, hold the **FLT2** terminal at low level. With this configuration, it is possible to recognize the result of error detection based on the logic level at the **FLT2** terminal and whether it is oscillated.

(286) FIG. **22** is a diagram showing a third example of sharing a fault signal output terminal (here, the **FLT2** terminal). As shown there, the fault detector-controller **442** can, if the adjustment data **D1** and **D2** is normal, hold the **FLT2** terminal at high level; if a one-bit error is detected in the adjustment data **D1** and **D2**, pulse-drive (oscillate) the **FLT2** terminal at a pulse frequency **f1**; and if a two-or-more-bit error is detected in the adjustment data **D1** and **D2**, pulse-drive the **FLT2** terminal at a pulse frequency **f2** (e.g., $f2 > f1$). With this configuration, it is possible to recognize the result of error detection based on the logic level and the pulse frequency at the **FLT2** terminal.

(287) While no more illustration will be given, a fault signal output terminal can be shared in any manner other than as described above, by changing the output state at the **FLT2** terminal as desired according to the result of error detection with respect to the adjustment data **D1** and **D2**. For example, a parameter (such as pulse width, duty, or type of modulation) with which the **FLT2** terminal is pulse-driven can be changed.

Application to Electric Vehicles

(288) FIG. **23** is a diagram showing the exterior appearance of an electric vehicle. As has been described above, the isolated gate driver **1** described previously and the traction inverter **400** employing it can be used suitably as a means for driving a motor on an electric vehicle **X10**.

Overview

(289) To follow is an overview of the various embodiments described above.

(290) For example, according to what is disclosed herein, an isolated gate driver includes: a switch connection terminal configured to have a switching element externally connected to it; a non-volatile memory having adjustment data written to it; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive the gate of the switching element with a characteristic set based on a value stored in the register; and a control logic circuit configured to keep the gate driving circuit in a disabled state until completion of reading the adjustment data from the non-volatile memory and storing the adjustment data in the register. (A first configuration.)

(291) For another example, according to what is disclosed herein, an isolated gate driver includes: a

switch connection terminal configured to have a switching element externally connected to it; a memory connection terminal configured to have externally connected to it a non-volatile memory having adjustment data written to it; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive the gate of the switching element with a characteristic set based on a value stored in the register; and a control logic circuit configured to keep the gate driving circuit in a disabled state until completion of reading the adjustment data from the non-volatile memory and storing the adjustment data in the register. (A second configuration.)

(292) The isolated gate driver of the first or second configuration described above may further include: a fault detector configured to detect a fault with the characteristic set based on the value stored in the register. (A third configuration.)

(293) In the isolated gate driver of any of the first to third configurations described above, the control logic circuit may be configured to start reading the adjustment data from the non-volatile memory after, as the supply voltage fed to the isolated gate driver rises, UVLO is canceled; continues storing the adjustment data in the register while keeping the gate driving circuit in the disabled state until completion of storing the adjustment data in the register; and cancel the disabled state of the gate driving circuit on completion of storing the adjustment data in the register. (A fourth configuration.)

(294) In the isolated gate driver of any of the first, third, and fourth configurations described above, the following may be sealed in a single package: a first semiconductor chip having the circuit elements of a primary circuit system integrated in it; a second semiconductor chip having the circuit elements of a secondary circuit system integrated in it; and a third semiconductor chip having integrated in it an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips. (A fifth configuration.)

(295) In the isolated gate driver of the second configuration described above, the following may be sealed in a single package: a first semiconductor chip having the circuit elements of a primary circuit system integrated in it; a second semiconductor chip having the circuit elements of a secondary circuit system integrated in it; and a third semiconductor chip having integrated in it an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips. (A sixth configuration.)

(296) In the isolated gate driver of the fifth or sixth configuration described above, the register and the gate driving circuit may both be integrated in the second semiconductor chip. (A seventh configuration.)

(297) For another example, according to what is disclosed herein, a traction inverter includes: the isolated gate driver of the first configuration described above; and the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have its gate driven by the isolated gate driver. (An eighth configuration.)

(298) For another example, according to what is disclosed herein, a traction inverter includes: the isolated gate driver of the second configuration described above; the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have its gate driven by the isolated gate driver; and the non-volatile memory externally connected to the memory connection terminal of the isolated gate driver, the non-volatile memory being configured to store the adjustment data for the isolated gate driver. (A ninth configuration.)

(299) For another example, according to what is disclosed herein, a traction inverter includes: a plurality of the isolated gate drivers of the sixth configuration described above; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have their gates driven by the plurality of the isolated gate drivers respectively; and a plurality of the non-volatile memories externally

connected to a plurality of the memory connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the non-volatile memories being configured to store the adjustment data for the plurality of the isolated gate drivers respectively. (A tenth configuration.)

(300) For another example, according to what is disclosed herein, a traction inverter includes: a plurality of the isolated gate drivers of the sixth configuration described above; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have their gates driven by the plurality of the isolated gate drivers respectively; and a single non-volatile memory as the non-volatile memory externally connected to all of a plurality of the memory connection terminals provided at the first-semiconductor-chip side of the plurality of the isolated gate drivers, the non-volatile memory being configured to store the adjustment data for all of the plurality of the isolated gate drivers. (An eleventh configuration.)

(301) For another example, according to what is disclosed herein, an electric vehicle includes: the traction inverter of any of the eighth to eleventh configurations described above. (A twelfth configuration.)

(302) For another example, according to what is disclosed herein, an isolated gate driver includes: a switch connection terminal configured to have a switching element externally connected to it; a non-volatile memory having adjustment data written to it; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive the gate of the switching element with a characteristic set based on a value stored in the register; a fault detector configured to perform fault detection with respect to other than the non-volatile memory; an error detection-correction circuit configured to perform error detection and error correction with respect to the adjustment data written to the non-volatile memory; a first external terminal configured to externally output the result of the fault detection; a second external terminal configured to externally output the result of the error detection; and a fault controller configured to, if a one-bit error is detected in the adjustment data, bring the second external terminal into an error-indicating output state and continue with normal operation of the gate driving circuit and, if a two-or-more-bit error is detected, forcibly stop the gate driving circuit. (A thirteenth configuration.)

(303) For another example, according to what is disclosed herein, an isolated gate driver includes: a switch connection terminal configured to have a switching element externally connected to it; a memory connection terminal configured to have externally connected to it a non-volatile memory having adjustment data written to it; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive the gate of the switching element with a characteristic set based on a value stored in the register; a fault detector configured to perform fault detection with respect to other than the non-volatile memory; an error detection-correction circuit configured to perform error detection and error correction with respect to the adjustment data written to the non-volatile memory; a first external terminal configured to externally output the result of the fault detection; a second external terminal configured to externally output the result of the error detection; and a fault controller configured to, if a one-bit error is detected in the adjustment data, bring the second external terminal into an error-indicating output state and continue with normal operation of the gate driving circuit and, if a two-or-more-bit error is detected, forcibly stop the gate driving circuit. (A fourteenth configuration.)

(304) In the isolated gate driver of the thirteenth or fourteenth configuration described above, the fault controller may be configured to, if a two-or-more-bit error is detected in the adjustment data, bring the first external terminal into a fault-indicating output state regardless of the result of the fault detection. (A fifteenth configuration.)

(305) In the isolated gate driver of any of the thirteenth to fifteenth configurations described above, the fault controller may be configured to, if a fault is detected with respect to other than the non-

volatile memory, bring the first external terminal into a fault-indicating output state regardless of the result of the error detection and forcibly stop the gate driving circuit. (A sixteenth configuration.)

(306) Or, in the isolated gate driver of the thirteenth or fourteenth configuration described above, the fault controller may be configured to, if a one-bit error is detected in the adjustment data, bring the second external terminal into a first output state and, if a two-or-more-bit error is detected in the adjustment data, bring the second external terminal into a second output state different from the first output state. (A seventeenth configuration.)

(307) In the isolated gate driver of any of the thirteenth and fifteenth to seventeenth configurations described above, the following may be sealed in a single package: a first semiconductor chip having circuit elements of a primary circuit system integrated in it; a second semiconductor chip having circuit elements of a secondary circuit system integrated in it; and a third semiconductor chip having integrated in it an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips. (A eighteenth configuration.)

(308) In the isolated gate driver of the fourteenth configuration described above, the following may be sealed in a single package: a first semiconductor chip having circuit elements of a primary circuit system integrated in it; a second semiconductor chip having circuit elements of a secondary circuit system integrated in it; and a third semiconductor chip having integrated in it an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips. (A nineteenth configuration.)

(309) In the isolated gate driver of the eighteenth to nineteenth configuration described above, the register and the gate driving circuit may both be integrated in the second semiconductor chip. (A twentieth configuration.)

(310) For another example, according to what is disclosed herein, a traction inverter includes: the isolated gate driver of the thirteenth configuration described above; and the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have its gate driven by the isolated gate driver. (A twenty-first configuration.)

(311) For another example, according to what is disclosed herein, a traction inverter includes: the isolated gate driver of the fourteenth configuration described above; the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have its gate driven by the isolated gate driver; and the non-volatile memory externally connected to the memory connection terminal of the isolated gate driver, the non-volatile memory being configured to store the adjustment data for the isolated gate driver. (A twenty-second configuration.)

(312) For another example, according to what is disclosed herein, a traction inverter includes: a plurality of the isolated gate drivers of the nineteenth configuration described above; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have their gates driven by the plurality of the isolated gate drivers respectively; and a plurality of the non-volatile memories externally connected to a plurality of the memory connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the non-volatile memories being configured to store the adjustment data for the plurality of the isolated gate drivers respectively. (A twenty-third configuration.)

(313) For another example, according to what is disclosed herein, a traction inverter includes: a plurality of the isolated gate drivers of the nineteenth configuration described above; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have their gates driven by

the plurality of the isolated gate drivers respectively; and a single non-volatile memory as the non-volatile memory externally connected to all of a plurality of the memory connection terminals provided at the first-semiconductor-chip side of the plurality of the isolated gate drivers, the non-volatile memory being configured to store the adjustment data for all of the plurality of the isolated gate drivers. (A twenty-fourth configuration.)

(314) For another example, according to what is disclosed herein, an electric vehicle includes: the traction inverter of any of the twenty-first to twenty-fourth configurations described above. (A twenty-fifth configuration.)

(315) According to the invention disclosed herein, it is possible to provide an isolated gate driver that contributes to size reduction and cost reduction of sets, and to provide traction inverters and electric vehicles that employ such isolated gate drivers.

Further Modifications

(316) The embodiments described above deal with examples where an isolated gate driver is applied to a traction inverter incorporated in an electric vehicle. This however is not meant to limit the application of isolated gate drivers, which find wide applications in industrial equipment, medical equipment, and the like.

(317) What is disclosed herein as to the implementation of a non-volatile memory and as to novel start-up sequences and fault output in that connection can be applied not only to isolated gate drivers but to signal transmission devices in general, regardless of whether they are of an insulated or non-insulated type.

(318) The various technical features disclosed herein may be implemented in any manners other than as in the embodiments described above, and allow for many modifications without departure from the spirit of their technical ingenuity. That is, the embodiments described above should be understood to be in every aspect illustrative and not restrictive, and the technical scope of the present invention is defined not by the description of the embodiments given above but by the appended claims and encompasses any modifications within a scope equivalent in significance to what is claimed.

Claims

1. An isolated gate driver, comprising: a switch connection terminal configured to have a switching element externally connected thereto; a non-volatile memory having adjustment data written thereto; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive a gate of the switching element with a characteristic set based on a value stored in the register; and a control logic circuit configured to keep the gate driving circuit in a disabled state until completion of reading the adjustment data from the non-volatile memory and storing the adjustment data in the register.
2. The isolated gate driver according to claim 1, further comprising: a fault detector configured to detect a fault with the characteristic set based on the value stored in the register.
3. The isolated gate driver according to claim 1, wherein the control logic circuit is configured to start reading the adjustment data from the non-volatile memory after, as a supply voltage fed to the isolated gate driver rises, UVLO is canceled, continues storing the adjustment data in the register while keeping the gate driving circuit in the disabled state until completion of storing the adjustment data in the register, and cancel the disabled state of the gate driving circuit on completion of storing the adjustment data in the register.
4. The isolated gate driver according to claim 1, wherein a first semiconductor chip having circuit elements of a primary circuit system integrated therein, a second semiconductor chip having circuit elements of a secondary circuit system integrated therein, and a third semiconductor chip having integrated therein an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips are sealed in a single package.

5. The isolated gate driver according to claim 4, wherein the register and the gate driving circuit are both integrated in the second semiconductor chip.
 6. A traction inverter, comprising: the isolated gate driver according to claim 1; and the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have a gate thereof driven by the isolated gate driver.
 7. An electric vehicle, comprising: the traction inverter according to claim 6.
 8. An isolated gate driver, comprising: a switch connection terminal configured to have a switching element externally connected thereto; a memory connection terminal configured to have externally connected thereto a non-volatile memory having adjustment data written thereto; a register configured to store the adjustment data read from the non-volatile memory; a gate driving circuit configured to drive a gate of the switching element with a characteristic set based on a value stored in the register; and a control logic circuit configured to keep the gate driving circuit in a disabled state until completion of reading the adjustment data from the non-volatile memory and storing the adjustment data in the register.
 9. The isolated gate driver according to claim 8, wherein a first semiconductor chip having circuit elements of a primary circuit system integrated therein, a second semiconductor chip having circuit elements of a secondary circuit system integrated therein, and a third semiconductor chip having integrated therein an insulating element for transmitting a signal between, while isolating between, the first and second semiconductor chips are sealed in a single package.
 10. A traction inverter, comprising: the isolated gate driver according to claim 8; the switching element externally connected to the switch connection terminal of the isolated gate driver, the switching element being configured to have a gate thereof driven by the isolated gate driver; and the non-volatile memory externally connected to the memory connection terminal of the isolated gate driver, the non-volatile memory being configured to store the adjustment data for the isolated gate driver.
 11. A traction inverter, comprising: a plurality of the isolated gate drivers according to claim 9; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at a second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have gates thereof driven by the plurality of the isolated gate drivers respectively; and a plurality of the non-volatile memories externally connected to a plurality of the memory connection terminals provided at the second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the non-volatile memories being configured to store the adjustment data for the plurality of the isolated gate drivers respectively.
 12. A traction inverter, comprising: a plurality of the isolated gate drivers according to claim 9; a plurality of the switching elements externally connected to a plurality of the switch connection terminals provided at a second-semiconductor-chip side of the plurality of the isolated gate drivers respectively, the plurality of the switching elements being configured to have gates thereof driven by the plurality of the isolated gate drivers respectively; and a single non-volatile memory as the non-volatile memory externally connected to all of a plurality of the memory connection terminals provided at a first-semiconductor-chip side of the plurality of the isolated gate drivers, the non-volatile memory being configured to store the adjustment data for all of the plurality of the isolated gate drivers.
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